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HOW IS RESEARCH POLICY ACROSS THE OECD ORGANISED?

INSIGHTS FROM A NEW POLICY DATABASE

OECD SCIENCE, TECHNOLOGY AND INDUSTRY
POLICY PAPERS
October 2018 **No. 55**

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HOW IS RESEARCH POLICY ACROSS THE OECD ORGANISED? INSIGHTS FROM A NEW POLICY DATABASE

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ABSTRACT

Building on a newly created policy indicator database, this paper provides a first systematic comparison of the governance of public research policy across 35 OECD countries from 2005 to 2017. The database was obtained following a three-year process that involved the development of an ontology of the governance of public research policy as well as data collection and validation by national authorities. The data show diverse institutions and mechanisms of policy action regarding higher education institutions (HEIs) and public research institutes (PRIs) are in place across the 35 OECD countries. The data also shows an increasing use of project funding, performance contracts and performance evaluations for HEIs and PRIs. In many countries, HEIs and PRIs are autonomous regarding their relations with industry, budget allocation but less frequently regarding salaries. Recent reforms have strengthened external stakeholders' participation in their governance, reflecting efforts to open research institutions.

The database is publicly available on the following webpage: https://stip.oecd.org/resgov.

Keywords: Innovation and research policy, higher education institutions (HEIs) and public research institutes (PRIs), governance of research policy, cross-country policy indicators, OECD countries

JEL codes: H11, I23, I28, O38

Acknowledgements: The authors are grateful to Cynthia Lavison for research assistance as well as to Andrés Barreneche, Diogo Machado, Evgeny Moiseichev, Tadanori Moriguchi, Sandra Planes Satorra, Akira Tachibana, and Malte Tötzke for help with the data collection. Detailed feedback on the policy questionnaire from experts and delegates of the OECD Working Party on Innovation and Technology Policy (TIP) and the OECD Committee for Scientific and Technological Policy (CSTP) are gratefully acknowledged. The authors would also like to thank Dominique Guellec, Michael Keenan, Philippe Larrue, and Blandine Serve for their insightful comments and Sylvain Fraccola for his help in building the database website.

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KEY FINDINGS

1. Institutions in charge of priority setting, budget allocation and evaluation

The following observations can be made regarding the institutions in charge of setting key priorities, budget allocations and evaluations:

National ministries and agencies are in charge of setting key policy priorities for HEIs and PRIs policy. In 11 of 34 countries (or 32%), a single ministry decides on research and innovation agendas, while there are separate ministries for research and innovation in 6 of 34 countries (or 18%) and 2 regions in Belgium. Agencies, which are public entities in charge of policy implementation regarding public research, or research and innovation councils, i.e. public institutions outside ministries with the mandate to support the governance of public research, set policy priorities in 11 of 34 (32%) countries.

National agencies decide on allocations of project-based funding for HEIs in 31 (or 89%) of 35 OECD countries, while in most countries ministries provide institutional block funding (32 of 35 for HEIs, 27 of 34 for PRIs). 9 countries have created agencies to allocate project-based funding between 2005 and 2017, such as the National Research Agency (ANR) in France (created in 2006), the Innovation Fund Denmark (IFD) (created in 2014), or the Spanish Agency of Research (AEI) (created in 2015).

Funding arrangements differ: While in 12 of 31 OECD countries where agencies provide project-based funding there is a single agency, the remaining 19 have more than one agency. These agencies are specialised in specific research fields in 2 out of 19 countries with agencies (or 11%) (i.e. in Australia, Canada) or cover separately research and innovation tasks (12 of 19 or 63%). In 5 countries (or 26%), agencies that cover separately research and innovation tasks and agencies specialised in specific research fields co-exist. Over 2007-17, several countries including Denmark and Estonia have reduced the number of funding agencies, in some cases to simplify funding applications (creating a "one-stop-shop"), to increase efficiency and in others to reduce the fragmentation of funding.

Agencies specialised in evaluating and monitoring the performance of HEIs and PRIs are also increasingly important. 19 of 34 countries (59%) and Wallonia in Belgium have such agencies. Examples of recently established agencies and independent committees for evaluation and monitoring include the Agency for Assessment and Accreditation of Higher Education (A3ES) in Portugal (2007), the Independent Review Committee in the Netherlands (2012) and the National Agency for Evaluation of Universities and Research Institutes (ANVUR) in Italy in 2010. In most cases the same agency is not in charge of allocation of funding and evaluation of performance.

Performance contracts and performance-based funding instruments have gained in importance, illustrating the move towards stronger performance and evaluation. Performance contracts that bind future funding of HEIs and PRIs to pre-defined targets are in place in 13 of 35 (37%) OECD countries and in a number of regions/federal states, (e.g. Scotland in the UK; Baden Wurttemberg, Brandenburg, and North-Rhine Westphalia among other federal states in Germany). In 9 countries, performance contracts were introduced in the past decade. Another 7 countries that do not have performance contracts introduced performance indicators in the formula that is used to allocate institutional funding to HEIs.

2. Policy co-ordination mechanisms

The following observations can be made on different national approaches to co-ordinate policy action regarding HEIs and PRIs:

Councils for research and innovation (referred to as councils from this section onwards) **are common across the OECD**. Councils are public institutions outside of ministries and agencies and whose mandate includes priority setting, policy advice and policy coordination regarding HEIs and PRIs. They are in place in 31 of 35 (89%) OECD countries and 15 were created since 2007.

Councils have different mandates. In 28 of 31 countries (90%) with councils, the councils provide policy advice and for 8 of them, those of Australia, Belgium, Canada, Denmark, Germany, the Netherlands, Portugal, and Sweden, this is their core role. In 23 of 31 countries with councils (or 74%), the council also develops national strategic priorities and in 15 of 31 (48%) councils, the councils evaluate policy reforms. Councils also differ with regards to their staff and budget and the stakeholders involved (see below).

Science, technology, and innovation (STI) strategies or plans are in place in most countries (33 of 35, 94%). These commonly define STI strategies to address major societal challenges (30 of 33, 91%). Key themes include sustainable growth, health, and efficient transportation systems. STI strategies and plans also define specific scientific research, technologies or economic fields of national priority (31 of 33, 94%). In 23 of 32 countries (72%), STI strategies address specific sub-national priorities for specific federal states or regions, reflecting for EU member states and partner countries Smart Specialisation strategies.

National STI strategies often include quantifiable targets. In 21 of 33 countries (61%) and in the Brussels-Capital Region, the Flemish Region and the Walloon Region in Belgium, national STI strategies include such quantitative targets. All 21 countries with quantitative targets and the Brussels-Capital Region, the Flemish Region and the Walloon Region in Belgium list R&D spending targets. Other targets include increasing funding for doctoral students (7 of 33 and the Brussels-Capital Region in Belgium) or job placements of researchers and PhDs in industry (5 of 33 and the Brussels-Capital Region in Belgium).

3. External stakeholder involvement and autonomy of HEIs and PRIs

The following observations can be made on different national approaches to the autonomy of HEIs and PRIs and external stakeholder involvement:

Regarding external stakeholder consultations in councils, private sector representation is widespread in research policy councils. While civil society – including members of labour unions and non-profit organisations (NGOs) - are active in 15 of 31 (or 48%) of councils, private sector representatives – often large firms but also in some cases SMEs - are present in 26 councils (or 84%).

Foreign experts participate in 6 of 31 (19%) OECD countries with councils (Austria, France, Germany, Greece, Switzerland, and the United Kingdom). Foreign experts are mostly from academia and, in a few cases, including Austria and the United Kingdom, from industry or the public sector.

External stakeholder involvement at the level of HEIs is also important, particularly with increased autonomy of HEIs and PRIs. University boards have stakeholder representation from outside universities in 28 of 34 countries (82%). 25 of these 28 (or

90%) countries have private sector representatives and 23 of 28 (82%) countries have civil society representatives. In Portugal and France, external stakeholder participation was introduced over the past decade (in 2007 and 2013 respectively).

In a large number of countries HEIs and PRIs are autonomous regarding industry relations, budget allocation, recruitment, and promotions of researchers. In 29 of 34 (85%) OECD countries, HEIs are free to create legal entities, such as technology offices and spinoffs, and decide on the conditions of collaboration with industry. In 23 of 34 (68%) countries, HEIs are free to decide on how they allocate their block funding internally.

Restrictions to autonomy apply mostly for salaries. HEIs can decide on the salaries of their academic staff in 12 of 34 (35%) OECD countries. In some countries, national laws regulate salary bands for academic personnel (e.g. Denmark and France), while in others collective bargaining agreements are in place at the national and industry level (e.g. Austria and the Netherlands).

ABBREVIATIONS AND ACRONYMS

Country and region abbreviations

AUS	Australia
AUT	Austria
BEL	Belgium
BEL-BC	Brussels-Capital Region (Belgium)
BEL-FL	Flemish Region (Belgium)
BEL-FR	French Community (Belgium)
BEL-WA	Walloon Region (Belgium)
CAN	Canada
CHL	Chile
CZE	Czech Republic
DNK	Denmark
EST	Estonia
FIN	Finland
FRA	France
DEU	Germany
DEU-BB	Brandenburg (Germany)
DEU-BW	Baden Württemberg (Germany)
DEU-NW	North Rhine-Westphalia (Germany)
GRC	Greece
HUN	Hungary
ISL	Iceland
IRL	Ireland
ISR	Israel
ITA	Italy
JPN	Japan
KOR	Korea
LVA	Latvia
LUX	Luxembourg
MEX	Mexico
NLD	Netherlands
NZL	New Zealand
NOR	Norway
POL	Poland
PRT	Portugal
SVK	Slovak Republic
SVN	Slovenia
ESP	Spain
SWE	Sweden
CHE	Switzerland
TUR	Turkey
GBR	United Kingdom
USA	United States
USA-CA	California (United States)
USA-MA	Massachusetts (United States)

Other abbreviations and acronyms

AEI	State Research Agency (Agencia Estatal de Invesitgación) (Spain)
ANR	National Research Agency (l'Agence Nationale de la Recherche) (France)
ANVUR	National Agency for the Evaluation of Universities and Research Institutes (l'Agenzia Nazionale di Valutazione del Sistema Universitario e della Ricerca) (Italy)
AOI	Agency for Entrepreneurship and Innovation (Belgium)
AQ	Agency for Quality Assurance and Accreditation Austria (Agentur für Qualitätssicherung und Akkreditierung Austria)
BMWFW	Federal Ministry of Science, Research and Economics (Bundesministerium für Wissenschaft, Forschung und Wirtschaft) (Austria)
CDTI	Centre for the Development of Industrial Technology (Centro para el Desarollo Tecnológico Industrial) (Spain)
CIHR	Canadian Institutes of Health Research
CNRS	National Center for Scientific Research (Centre National de la Recherche Scientifique) (France)
CPCTI	Council, the Council for Scientific, Technological and Innovation Policies (Consejo de Política Científica, Tecnológica y de Innovación) (Spain)
CRDI	Council for Research, Development and Innovation (Czech Republic)
CSEM	Swiss Center for Electronics and Microtechnology (Centre Suisse d'Electronique et de Microtechnique)
CSIC	National Research Council (Consejo Superior de Investigaciones Científicas) (Spain)
CSTI	Council for Science, Technology and Innovation (Japan)
DFG	German Research Foundation (Deutsche Forschungsgemeinschaft)
DLO	Agricultural Research Service (Dienst Landbouwkundig Onderzoek) (Netherlands)
EC	European Commission
ECN	Energy Centre of the Netherlands (Energieonderzoek Centrum Nederland)
ERA	European Research Area
ERC	European Research Council
ERDF	European Regional Development Fund
EU	European Union
FCT	Foundation for Science and Technology (Fundação para a Ciência e a Tecnologia) (Portugal)
FFG	Research Promotion Agency (Forschungförderungsgesellschaft) (Austria)
FWF	Austrian Science Fund (Fonds zur Förderung der Wissenschaft und Forschung)
FWO	Research Foundation Flanders (Fonds Wetenschappelijk Onderzoek) (Belgium)
HCERES	High Council for the Evaluation of Research and Higher Education (Le Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur) (France)
HEA	Higher Education Authority (Ireland)
HEI	Higher education institution
ICT	Information and communication technologies
IFD	Innovation Fund Denmark (Innovationsfonden)
ISCIII	Institute of Health Carlos III (Instituto de Salud Carlos III) (Spain)
NGO	Non-governmental organisation
Marin	Maritime Research Institute of the Netherlands (Maritiem Research Instituut Nederland)

MBIE	Ministry of Business, Innovation and Employment (New Zealand)
MEC	Ministry of Education and Culture (Suomen opetus- ja kulttuuriministeriö) (Finland)
NHMRC	National Health and Medical Research Council (Australia)
NLR	National Aerospace Laboratory (Nationaal Lucht- en Ruimtevaartlaboratorium) (Netherlands)
NRP	National Reform Programme
NSERC	Natural Sciences and Engineering Research Council (Canada)
NSF	National Science Foundation (United States)
PRI	Public research institution
QQI	Quality and Qualifications Ireland
RIS	Research and Innovation Strategy for Smart Specialisation
SciSIP	Science of Science and Innovation Policy (United States)
SIO	Strategic Innovation Areas (Strategiska Innovationsområden) (Sweden)
SME	Small and medium-sized enterprise
SoSP	Interagency Working Group on the Science of Science Policy (United States)
SSC	Swiss Science Council (Schweizerischer Wissenschaftsrat)
SSHRC	Social Sciences and Humanities Research Council of Canada
STEM	Science, technology, engineering and mathematics
STI	Science, Technology and Innovation
TNO	Netherlands Organisation for Applied Scientific Research (Nederlandse Organisatie voor Toegepast-Natuurwetenschappelijk Onderzoek)
TO2	Applied Research Institutions (Toegepast Onderzoek Organisaties) (Netherlands)
TÜBİTAK	Scientific and Technological Research Council of Turkey
UPRO	Umbrella Public Research Organisation
VINNOVA	Government Agency for Innovation Systems (Verket för Innovationssystem) (Sweden)

INTRODUCTION

The contributions of research conducted by universities (or higher education institutions, HEIs) and public research institutions (PRIs) to innovation are well recognised as is the need for public support for such research. To avoid underinvestment in public research, a battery of policy instruments is deployed not only on finance but also optimise the contributions of HEIs and PRIs to the economy and society. These policy mixes are shaped by a set of policy mechanisms and institutional arrangements that are quite different across countries, including the degree of autonomy of universities, the role of councils in charge of research and innovation, STI strategies as well as the type of funding arrangements of HEIs and PRIs. Together, policy mechanism and institutions constitute the governance system surrounding HEIs and PRIs. Continued policy support for publicly-funded research at HEIs and PRIs require cross-country evidence on governance of public research. Yet, systematic comparisons across countries were limited to qualitative comparisons of selected characteristics or selected countries, rendering an understanding of common trends and differences impossible.

This paper provides a first systematic mapping of the governance of publicly-funded research conducted in higher research institutions and public research institutions across all 35 OECD countries for the period 2005 to 2017 using a new policy database. The mapping builds on an ontology developed to compare countries with regards to how various governmental bodies (ministries, councils, and agencies), HEIs, PRIs and private actors are involved in the design and implementation of policies regarding publicly-funded research conducted in HEIs and PRIs. Three dimensions are covered: i) the institutions in charge of setting scientific and thematic priorities, the allocation of public funding and funding arrangements (such as performance contracts) as well as the monitoring and evaluation of HEI/PRI performance; ii) mechanisms to co-ordinate policies aimed at supporting the contributions of HEIs and PRIs to innovation, including research and innovation councils, national research and innovation strategies or plans and inter-agency joint programming; iii) external stakeholder involvement in policy making, including HEIs and PRIs themselves and their autonomy as well as involvement of the private sector, NGOs among others.

The paper describes a new database that was created following a three-year process that involved the development of an ontology of governance of public research and in-depth data collection and validation. Information on country policies was collected using a policy survey and the data quality was validated in interaction with country delegates and experts of the OECD Committee of Science and Technology Policy (CSTP) and the OECD Technology and Innovation Policy (TIP) Working Party between March 2017 and May 2018. The sources used include the European Commission/OECD International Survey on Science, Technology and Innovation Policies and the OECD Country Innovation Policy Reviews. Where additional information was needed, national legislative texts, strategic policy documents, and other such official documents (statements on the status of councils, committees, etc.) were used.

The evidence collected shows that diverse mechanisms of policy action regarding HEIs and PRIs across countries are in place. The data across the 35 OECD countries shows an increasing use of project funding, performance contracts and evaluations of performance for HEIs and PRIs. In many countries, HEIs and PRIs are autonomous regarding industry relations, budget allocation (including research funds) but less so regarding salaries. Recent reforms have strengthened external stakeholder participation, reflecting efforts to open research institutions. The database is publicly available for policy analysis on the following webpage: https://stip.oecd.org/resgov.

This paper contributes to the wider set of projects that develop cross-country indicators of policies for systematic comparison. This includes the World Bank's Doing Business (World Bank, 2018) and the OECD's Product Market Regulation (Koske et al, 2015) and Services-trade Restrictiveness indicators (Grosso et al., 2015). An earlier example for patent protection is the work by Ginarte and Park (1997) with a data update provided by Park (2008). This paper is a first attempt at providing such indicators within the set of research and innovation policies. A notable exception is the European University Autonomy survey of the European University Association (Estermann, Nokkala, and Steinel, 2015) that describes systematically for 29 European countries the degree autonomy of universities. The EC-OECD STIP survey (OECD, 2018) is a step forward with regards to providing policy indicators for the wider set of STI policies once relevant ontologies have been developed.

This paper also builds on a rich literature that has discussed the governance of public research. This includes in particular studies that have investigated the institutional arrangements that shape the design and implementation of the innovation policy mix from a cross-country perspective (e.g. Borrás and Edquist 2013; Borras and Edler, 2014). Other studies have focused on in-depth analyses of specific governance dimensions, such as the organisation of funding agencies and the type of funding arrangements in the United States (e.g. Breznitz, 2007; Fuchs, 2010). In the absence of data, there is little cross-country analysis of the impacts of differences in governance. An exception is the study of Aghion et al. (2010) that uses data on 197 European universities in 2006 to investigate the role of university autonomy for university performance.

This large-scale cross-country comparison has important limitations. A number of factors cannot be captured by this exercise, including the role of individuals in key position and institutional practice that may differ from roles as set out in laws and regulations. Moreover, while every effort was made to avoid possible ambiguities with regard to the terms used for the ontology, the diversity of country conditions has not in all cases allowed avoiding them for all countries. Challenges apply in particular to countries with strong sub-national governance structures of public research. Where this is the case, information on regional structures is also provided.

Rather than aiming at a full representation of country practice, these indicators offer a rough characterisation that allows for more systematic analysis of key aspects of governance across countries and their impact. This is valuable to develop policy advice as it is well-known that "one size does not fit all". Avenues for future research on the basis of this evidence include identifying cluster of countries characterised by similar governance practices and understanding how these may fit country characteristics relevant for public research. Regression-based approaches, as applied widely to the policy indicators described above (as e.g. Ginarte and Park, 1997 and Park, 2008) can also be applied, using either specific or combinations of the indicators described here.

The remainder of the paper is structured as follows. Section 1 provides findings on institutions in charge of priority setting, budget allocation and evaluation across the OECD, while Section 2 provides an overview of policy co-ordination arrangements and Section 3 describes different stakeholder consultation arrangements and levels of institutional autonomy. Section 4 concludes. The annex discusses the methodology to develop the framework and the data gathering process. It also presents the ontology that was developed.

1. INSTITUTIONS IN CHARGE OF PRIORITY SETTING, BUDGET ALLOCATION AND EVALUATION

This section describes institutions that take decisions on national scientific priorities (1.1), funding of research and innovation (1.2) including performance contracts for HEIs (1.3), and evaluation of HEI and PRI performance (1.4).

1.1. Setting of scientific priorities

Different institutions are in charge of setting policy priorities for HEIs and PRIs across the OECD, including national Ministries of Education, Research, and/or Innovation as well as specialised agencies, which are public institutions in charge of policy implementation, and councils, which are public institutions outside of ministries and agencies with a mandate to steer research policy. Data shows that the same institution generally sets priorities for both HEIs and PRIs (27 of 34 or 79% of countries).[1]

Most commonly, Ministries of Education, Research, and/or Innovation take decisions on national scientific and thematic priorities for HEIs and PRIs (Figure 1). Funding of HEIs and PRIs is used as a criterion to identify who is in charge of setting scientific, sectoral and/or thematic priorities. To illustrate the decision criterion: If competitive funding from an agency represents more than 50% of university funds and if the funding agency allocates funds to pre-determined themes, scientific fields or sectors that it has set itself, then the agency is identified as the actor who takes decisions on scientific, sectoral and/or thematic priorities. In 11 of 34 countries (or 32%), there is a single ministry that decides on research and innovation agendas for HEIs, while more than one ministries decide research and innovation agendas in 6 of 34 countries (or 18%) and in 2 Belgian regions (Austria, Belgium-Flemish Community, Belgium-French Community, Finland, Italy, Poland, Portugal, and the Slovak Republic). Agencies, which are public entities in charge of policy implementation regarding public research, set policy priorities for HEIs in 4 of 34 countries (or 12%) and the Capital Region Brussels in Belgium (32%). Research and innovation councils, i.e. public institutions outside ministries with the mandate to support the governance of public research, set policy priorities for HEIs in 7 of 34 countries (or 21%)

Federal systems have decentralised governance structures. In Belgium, Germany, Spain, Switzerland and the United States, regional or federal state ministries and agencies take decisions on national or regional scientific and thematic priorities. In Spain, regional authorities (Comunidades Autónomas) set priorities for regional PRIs. The Spanish National Research Council, the main national-level PRIs in Spain, decides itself on its scientific and thematic priorities. In Switzerland, Cantonal ministries set priorities for PRIs in coordination with the institution such as e.g. the Swiss Center for Electronics and Microtechnology (CSEM). In Belgium, research and innovation agendas that apply for HEIs and PRIs are set by the Department of Economy, Science and Innovation of the Flemish Community, the Ministry for Higher Education, Research and Medias along with the Directorate General for Non-Compulsory Education and Scientific Research (DGENORS) of the French Community, the Operational Directorate General for the Economy, Employment and Research (DGO6) of the Walloon Region and Innoviris and the Institute for the Encouragement of Scientific Research and Innovation of Brussels of the Brussels-Capital Region. The Federal Government has competence for ten federal PRIs (EC, 2015, p. 1).

Figure 1. Number of ministries in charge of defining scientific and thematic priorities of HEIs and PRIs

Note: This figure corresponds to question 1.1.a and 1.1.b ("Who mainly decides on the scientific, sectoral and/or thematic priorities of HEIs and PRIs?"). Information on HEIs is missing for New Zealand. Information on PRIs is missing for Australia and New Zealand.

[1] In Denmark, the GTS-institutes (public research institutes) propose projects bottom up that the Danish Agency for Institutions and Educational Grants then evaluates and funds. However, the funding of the institutes requires the approval from the Ministry of Higher Education and Science.

[2] In Austria, the regional level sets priorities in some cases.

[3] In the Netherlands, the Ministry of Economic Affairs, Ministry of Defence, the Ministry of Infrastructure and Environment and the Ministry of Education, Culture and Science set the agenda except for PRIs, which have their priorities set at the national agency level. Regarding HEIs, institutions themselves also play an important role in agenda setting.

[4] In Belgium the Federal science policy office (BELSPO) sets priorities for federal PRIs, community and regional ministries and agencies set priorities for community PRIs and regional institutions.

[5] In Spain, the regional level sets priorities for PRIs of Autonomous Communities.

Interpretation of the figure: First row, second column shows that in 9 countries scientific priorities for both HEIs and PRIs are decided by a single national Ministry.

Even where the final decision on national research and innovation priorities is taken by ministries and agencies, decision-taking with regards to research and innovation policy is not necessarily a top-down process:

- In Sweden, for instance, university-industry consortia, the Strategic Innovation Areas (Strategiska Innovationsområden, SIOs) were established in 2012 to jointly set scientific, sectoral and thematic strategic priorities that guide research funding. Research organisations and industry form research consortia around research topics of strategic interest to them. Jointly, they apply for the status of a strategic innovation area which receives public funding. The consortia jointly apply for public funding and if it is granted, they themselves allocate public funds to individual research projects. SIO funds are managed by Sweden's innovation agency VINNOVA (OECD, 2016b). Its programme "Internet of Things", for instance, includes Uppsala University as leading institution and programme officer, large companies such as Ericsson, and SMEs such as Sigma Connectivity, and Teyi Services, and a number of other HEIs and actors from government and civil society (e.g. Teknikföretagen, the Royal Institute of Technology, Malmö University, and the Swedish Electronics Trade Association).

- In some countries national priorities for research and innovation account only for a share of funding. In Luxembourg, for instance, institutional block funding, which represents 77% of the university budget and 63% of PRIs' budget is not allocated on the basis of specific research agenda but of performance indicators (OECD, 2016c).

There are considerable differences with regard to PRIs governance arrangements. Different types of PRIs exist within countries (OECD, 2011). PRIs can be organised as associations of public research institutions (i.e. Umbrella Public Research Organisation, UPROs) that set their own research and innovation agendas within broadly defined fields of national research policy, such as e.g. health. This is the case in 4 of the 33 OECD countries (or 12%). The Centre National de la Recherche Scientifique CNRS in France, Consejo Superior de Investigaciones Científicas (CSIC) in Spain, the Fraunhofer-Society and the Max-Planck Society in Germany, and the Hungarian Academy of Sciences are such UPROs.

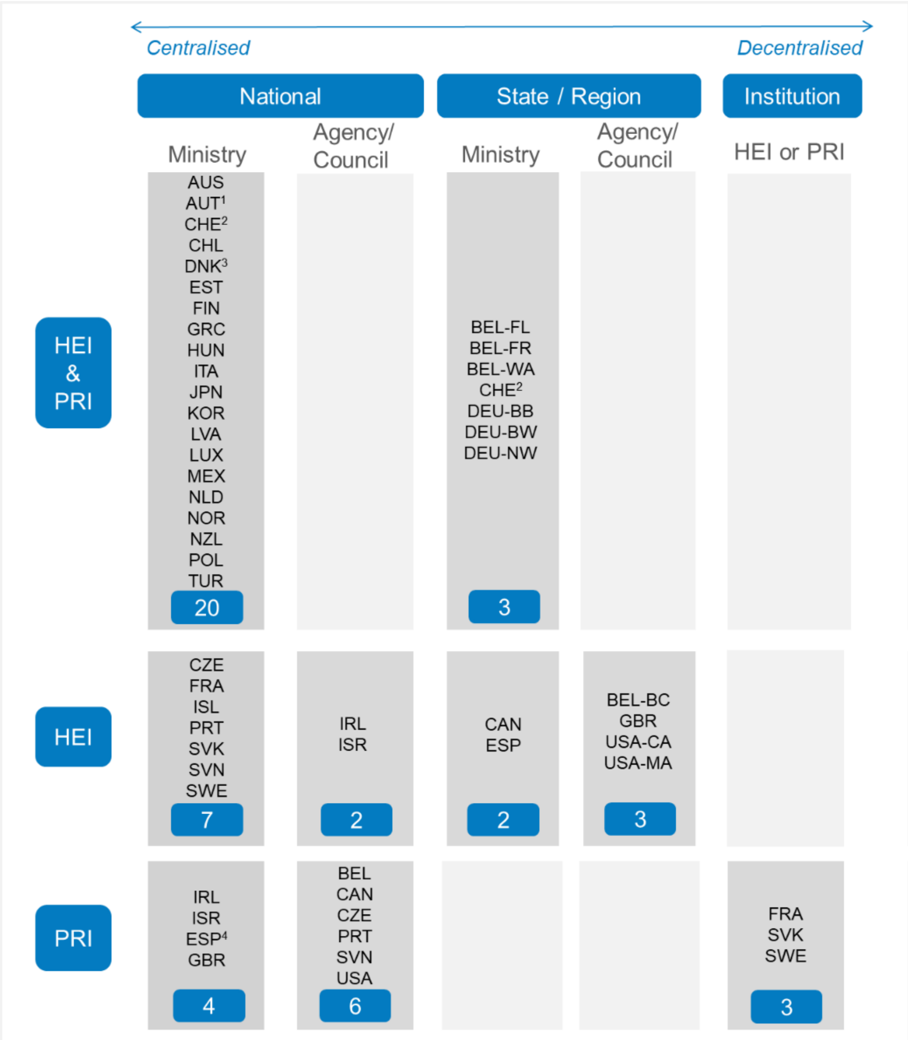
Other PRIs are government-owned or run by government departments or ministries at the national and sub-national levels. Often, national or regional ministries set scientific priorities for PRIs. In Finland, PRIs were established on a sectoral basis to serve the needs of sectoral ministries. Their role has traditionally been to undertake research in specific sectors but during the past 10 years their research has become more international and cross-sectoral. In Spain, scientific and thematic priorities of PRIs other than the CSIC are set out in regional-level science and technology plans, in their own mission statements or by the informal mandates set out by regional political authorities (OECD, 2011, p. 87).

1.2. Allocating budgets

1.2.1. Block funding

Institutional block funding is public funding allocated to HEIs or PRIs for their education, research and innovation activities with only general provisions as to how it is spent. Generally national ministries provide institutional block funding (block grants) in 32 of 35 OECD countries (or 91%) for HEIs, while they do so in 27 of the 34 OECD countries (or 79%) for PRIs[2] (Figure 2).

Figure 2. Institutions in charge of allocating block funding to HEIs and PRIs



Note: This figure corresponds to question 1.2.a and 1.2.b ("Who allocates institutional block funding to HEIs and PRIs?"). Information on PRIs is missing for Iceland.

[1] In Austria some institutions receive funding from the regional level.

[2] In Switzerland, the federal government is in charge of the supervision of the federal HEIs, while the cantons are in charge of the supervision of the cantonal universities.

[3] In Denmark, the GTS-institutes (public research institutes) propose projects bottom up that the Danish Agency for Institutions and Educational Grants then evaluates and funds. However, the funding of the institutes requires the approval from the Ministry of Higher Education and Science.

[4] In Spain, some regional institutions receive funding from the regional level.

Interpretation of the figure: First row, first column shows that in 20 countries national Ministries allocate institutional block funding to both HEIs and PRIs.

Funding arrangements for PRIs differ from those of HEIs. In 9 of 34 countries (26%), PRI funding is managed by dedicated agencies or PRIs themselves. In the Netherlands, responsibilities for PRI funding were changed in 2007 to strengthen the contributions of PRIs to the top sectors strategy (Box 1).

Box 1. PRI funding reforms in the Netherlands

In 2007, the Netherlands introduced a funding reform of their PRIs that aimed to better align research of PRIs to the demands of society and business. Before the reforms, the Ministry of Education, Culture and Science allocated block funding to PRIs which was not subject to sectoral priorities. After the reforms, funding requires PRIs to support the nine national strategic sectors (so-called "top sectors"): (1) horticulture and propagation materials, (2) agri-food, (3) water, (4) life sciences and health, (5) chemicals, (6) high tech, (7) energy, (8) logistics, and (9) creative industries.

In order to better align strategic research of PRIs with the nine sector, the Applied Research Institutions (Toegepast Onderzoek Organisaties, TO2) were created in 2010 as a consortium of the six largest PRIs: The Netherlands Organisation for Applied Scientific Research (TNO), the Agricultural Research Service (DLO), Deltares, the Energy Centre of the Netherlands (ECN), the National Aerospace Laboratory (NLR), and the Maritime Research Institute of the Netherlands (Marin). The T02 consortium co-ordinates research activities under the strategic research sectors.

Source: OECD (2014b), "Innovation actors in the Netherlands" in OECD Reviews of Innovation Policy: Netherlands, OECD Publishing, Paris, pp. 164-167.

In federal systems, institutional block funding of HEIs is handled by Federal State (regional) governments. In Belgium, Canada, Germany, Spain, and Switzerland regional Ministries of Education, Research, and/or Innovation provide university funds. In Switzerland, Cantonal Departments of Education such as e.g. the Hochschulamt Kanton Zürich are responsible.[3] In the United Kingdom and the United States, Federal State (regional) agencies allocate block grants. In the United Kingdom, Higher Education Funding Councils exist in England, Scotland, and Wales. The United States have dedicated agencies at the level of Federal States that manage public HEIs funds, such as e.g. the Board of Higher Education in the State of Massachusetts.

1.2.2. Agencies and project-based funding

The other important component of HEI and PRI funding aside from block funding is project-based funding, i.e. which is funding that is attributed to a research group or a researcher to perform a specific research and/or innovation activity. In most countries project-based funding for HEIs and PRIs is managed by funding agencies (in 31 of 35 OECD countries, 89%). 10 of these institutions have been created after 2005. In France, for instance, the French National Research Agency (L'Agence nationale de la recherche, ANR) was created in 2006 to allocate project-based funding to the public research sector where such funding now represents 4% of total public budget for research in France (OECD, 2014a).

Ministries of Education, Research and/or Innovation continue to allocate project based funding in Greece, Italy, Japan, and New Zealand. In the case of New Zealand, the Ministry of Business, Innovation and Employment (MBIE) provides project-based funding following the merger of the Ministry of Research, Science and Technology and the funding agency Foundation for Research, Science and Technology in 2011.

The number of public funding agencies differs across countries. While in 12 of 31 OECD countries where agencies provide project-based funding there is a single agency in place, the remaining 19 have more than one agency (Figure 3). These agencies cover separately research and innovation tasks in 12 out of 19 countries (63%), or are specialised in specific research fields in 2 out of 19 countries with national agencies in place (or 11%). In 5 countries (or 26%), both agencies that cover separately research and innovation tasks and agencies that are specialised in specific research fields are in place. Specialised agencies also reflect federal governance structures as in the case of the Flemish-Region in Belgium.

Figure 3. Number of public agencies in charge of project-based funding allocations in 31 OECD countries with agencies in place

Note: The figure corresponds to question 1.3.c ("Name of the institution in charge of project-based funding"). Information is displayed for 31 countries for which project based funding is allocated by at least one national agency.

* The Swiss funding agency Innosuisse started operating only in 2018.

There are various reasons for a division of tasks between agencies. In Belgium, 5 regional funding agencies provide project-based funding due to the country's federal structure. In other countries there are agencies that are specialised in specific research fields. In Australia, the National Health and Medical Research Council (NHMRC) is specialised in managing funds for health and medical research while the Australian Research Council handles competitive calls for all other research fields. Canada has several of such specialised agencies, including the National Research Council, the Natural Sciences and Engineering Research Council (NSERC), the Canadian Institutes of Health Research (CIHR), and the Social Sciences and Humanities Research Council of Canada (SSHRC).

Agencies specialise also by type of research and innovation activity they fund. In Austria, Austrian Science Fund (FWF) is responsible for basic research while the Austrian Research Promotion Agency (FFG) and the CDG-Christian Doppler Research Association provides funding for applied research. In Flanders (Belgium), the Research Foundation Flanders (FWO) allocates project-based funding for research while the Agency for Entrepreneurship and Innovation (AOI) funds innovation activities. In Spain, the State Research Agency (AEI) is responsible for funding basic research, while the Centre for the Development of Industrial Technologies (CDTI) funds applied research and business innovation. Spain also has the Institute of Health Carlos III (ISCIII), which provides funding for health and biomedical research.

Denmark and Estonia have reduced the number of funding agencies over the period 2007-17. Single agencies were established, in some cases to simplify funding applications for institutions (creating a "one-stop-shop") and to reduce fragmentation of funding support, and in others to increase efficiency. Denmark created the Innovation Fund Denmark (IFD) in 2014 from the merger of the Danish Council for Strategic Research, the Danish National Advanced Technology Foundation and the Danish Council for Technology and Innovation. The objective of the merger was to simplify grant applications for researchers and businesses. In Estonia, the Estonian Research Council was created in 2012 as the single agency in charge of project-based funding by consolidating the functions of three agencies in order to reduce the fragmentation of public research funding.

In countries with federal structures, there is a division of tasks between national level and the Federal State or subnational-level with regard to education, on the one hand, and research and innovation, on the other hand. An exception is Belgium that has regional agencies in place. In Germany, for instance, project-based funds for research and innovation projects are allocated by the German Research Foundation (DFG) and partly from the Federal Ministry of Education and Research while universities receive their budgets from Federal State Ministries of Education. In Spain, while the Autonomous Communities (regional authorities) allocate block funding to public universities, the national-level State Agency for Research is the main source of project-based funding.

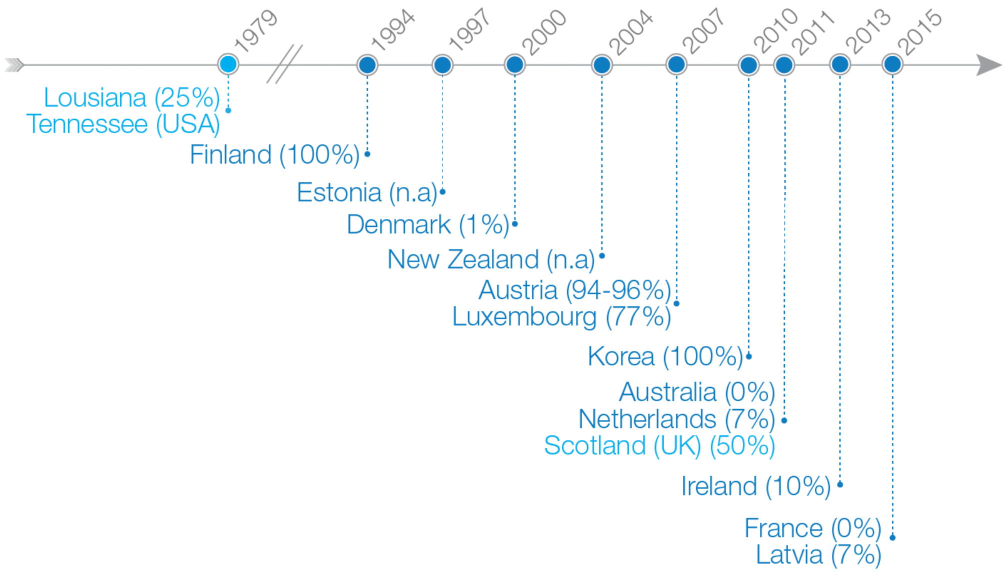
1.3. Performance contracts and other reforms to increase performance incentives

Performance contracts have gained in importance to reward performance of HEIs across the OECD. Performance contracts set performance targets for HEIs and in most cases bind a share or their block funding allocation. They are in place in 13 countries (37%, 13 of 35 OECD countries) and in a number of regions/federal states (Scotland in the UK; Louisiana and Tennessee in the USA; Baden Wurttemberg, Brandenburg, and North-Rhine Westphalia among other Federal States in Germany). Australia, Denmark and France have performance contracts in place that do not directly bind funding.

8 of the 13 OECD countries with performance contracts in place and Scotland in the United Kingdom have introduced them in the last decade (Figure 4). Australia (2011) and France (2015) introduced performance contracts that do not bind funding of universities. Slovenia is currently in the process of introducing performance contracts. Norway is currently rolling out a pilot exercise with 5 HEIs and aims to have a performance contracts system in place in 2019. Finland is an early adopter; the Ministry of Education and Culture (MEC) has had performance contracts with HEIs since 1994.

There are differences across countries on a number of aspects of performance contracts, including the shares of HEI budgets that are subject to performance contracts. Among the 9 countries and 4 regions/federal states for which such information was provided, the shares subject to performance contracts vary from very 0% in Australia and France, 1% in Denmark, 7% in Latvia and the Netherlands to much larger shares of 94-96% in Austria and 100% in Finland and Korea. At regional/federal level, in Scotland, for instance, performance contracts bind 50% of institutional funding of HEIs. In the case of German Federal States for which information is available, performance contracts apply to 2% of HEIs' institutional funding in the Federal State of Brandenburg and 23% of block funding of HEIs in North-Rhine Westphalia.

Figure 4. Year of introduction of performance contracts and shares of HEI institutional block funding involved*



Note: This figure corresponds to question 1.3.a ("Do performance contracts determine institutional block funding of HEIs?") and 1.3.b. ("Share of HEI budget subject to performance contracts"). Values in parentheses show the share of HEIs' institutional block funding subject to performance contracts. Information on the share of HEIs' budget subject to performance contracts is missing for Estonia, Japan, and New Zealand. At the regional/Federal State level, Scotland in the United Kingdom, Louisiana and Tennessee in the United States, and several German Federal States including Baden Wurttemberg, Brandenburg, and North-Rhine Westphalia have performance contracts in place.

* Performance contracts are in place in most Federal States in Germany, e.g. Brandenburg and North-Rhine Westphalia. Some Federal States introduced them in the 2000s while others had introduced them earlier. The share of HEI institutional block funding involved also varies, e.g. 2% in Brandenburg and 23% in North-Rhine Westphalia.

Other differences in performance contracts relate to the targets they define and the way they do so. Among the countries that provided information on targets, 12 countries and 2 regions (Scotland in the United Kingdom and the German state of North-Rhine Westphalia) included education and research performance indicators, such as target numbers for Master and PhD degrees to be awarded, the share of graduates employed one year after graduation, and the number of scientific publications in international journals.

In addition, 10 countries and 2 regions (Scotland in the United Kingdom and the German state of North-Rhine Westphalia) put emphasis on HEIs' role in support of business innovation. Innovation-related indicators include the number of patents filed by researchers and/or the university (Australia, Korea and Luxembourg), revenues from licencing intellectual property and contract research (Australia, Korea, Luxembourg, and Scotland in the United Kingdom), industry-funded R&D (Estonia, Finland, and North Rhine-Westphalia in Germany), the number of spin-offs of students and researchers (Australia, Denmark, Ireland, Korea, Luxembourg, and New Zealand), the number of collaborations with industry (Australia, Denmark and Ireland), and the number of innovation vouchers for particular science to business collaborations (Scotland in the United Kingdom).

Moreover, 5 countries (Australia, Denmark, France, Korea, and Luxembourg) and Scotland (UK) require HEIs to contribute to address societal challenges such as facilitating access of underrepresented groups to higher education and improving research contributions in support of sustainability. Australian and Scottish performance contracts for HEIs, for instance, target the number of students from underrepresented groups and from poor background. Performance contracts in Scotland state the goal to reduce energy emissions per student. Denmark asks HEIs to fund interdisciplinary research centres to tackle research of relevance to society. Topics include addressing growing inequalities, challenges brought by the digital transformation, and achieving more sustainable growth.

Finally, there are differences in how targets are defined. Some countries also specify a diversity of targets while others focus on a narrower set of targets. Australian, French, Korean, Luxembourgian and Scottish performance contracts, for instance, set targets regarding business innovation and support to the local economy and to addressing societal challenges. There are also differences in type of indicator. In some countries, qualitative indicators are used while others rely more on quantitative indicators.

Countries also differ with regard to the degree of involvement of HEIs in formulating performance targets. Table 1 describes the cases of Austria, Finland and Scotland (United Kingdom) as examples.

Performance contracts are relatively new policy instruments and have been subject to extensive policy experimentation. In Austria, performance contracts are in place since 2007 while several evaluation studies advocate for improving the current structure of performance agreements. The Austrian Science Council (2016) and De Boer et al. (2015) recommend using performance-based funding only for activities that support strengths of individual universities. Currently, the ability of performance contracts to strategically steer universities to specialise according to their strengths is limited because most funding is used by the institutions to cover their basic operations, especially personnel costs and existing infrastructure. Developments plans that are jointly created by universities and the Ministry set out strengths of each university such as e.g. research, teaching, or linkages to local industry. Stampfer (2017) argues that university budgets should be tied to concrete output activities instead of current lump sum allocations. Sanction mechanisms in cases where targets are not met would improve the steering function of performance contracts, while rewarding institutions that have met their targets.

Performance contracts are only one measured introduced over the past decade. Other reforms include the introduction of performance indicators in the formula for the allocation of university block grants. 7 countries introduced performance indicators since 2005 (Belgium, Czech Republic, Italy, Norway, Poland, Sweden, and Turkey).

Also, several countries strengthened programmes for research excellence. Germany, for instance, established the “Initiative for Excellence” in 2005. The Initiative for Excellence was a competition among German research universities for top-up funds from the Federal Government. It aimed to make German universities more competitive internationally. In 2007, three “excellence universities” were selected based on criteria of research excellence. Each university received USD 25.6 million (EUR 21 million) annually. Another 18 universities received funding to set up international graduate schools and “excellence clusters”, i.e. research hubs that bring together different research groups from within and across universities in the region. Its second round in 2012 expanded funding to 11 elite universities. In 2018, the Initiative for Excellence was continued under a new name, the “Excellence Strategy”, providing support only for existing excellence clusters and the selected excellence universities.

Table 1. Performance contracts in Austria, Finland and Scotland (United Kingdom)

Country	Targets	Process
Finland	Regarding education, quantitative indicators include ● the number of bachelor's, master's and PhD degrees awarded ● the percentage of students who have been awarded more than 55 study credits per academic year ● the number of employed graduates. For research, indicators include ● scientific publications ● the percentage of competitive funding in total funding of the institution. A number of indicators focus specifically on the degree of internationalisation, including ● the number of international teaching and research personnel ● the number of master's degrees awarded to foreign nationals ● student mobility to and from Finland. Indicators of other education and science policy considerations account for 28% and include strategic development efforts, field-specific funding and contributions to "national duties" (e.g. teacher training schools). A different formula applies for universities of applied science with criteria focusing on education (79%), R&D (15%) and strategic development (6%).	A funding formula serves as a basis for each university to negotiate its performance agreement with the Ministry of Education and Culture (MEC) at the beginning of every four-year term. Each performance agreement contains institution specific targets. Universities participate in the monitoring and evaluation process. The evaluation process also involves on-site visits by staff from the Ministry. Performance reviews are conducted jointly by representatives from the MEC and individual institutions. To enable the Ministry to monitor performance, HEIs provide information to a central statistical database that is maintained by MEC. Every year an assessment of HEIs' performance is published.
Austria	Qualitative and quantitative criteria used in performance contracts set targets for universities: education, research and innovation. Indicators for education include the number of students that complete full credits per academic year; the number of graduates; and the quality of teaching. Regarding universities' research role, specific attention is paid to the generation of basic research; as well as to the career paths of young academics. Regarding innovation outcomes, indicators vary across institutions. The University of Vienna, for instance, commits to increase the number of patents and to provide courses on technology transfer (University of Vienna, 2015).	Specific performance agreements are signed with each of the 22 Austrian institutions for a period of three years. They are based on institutional development plans. The National Development Plan for Higher Education formulated by the Federal Ministry of Science, Research and Economics (BMWFW) for a period of six years sets national objectives that informs the universities' development plans. Goals include progress in the number of students in different disciplines, improvement in the number of graduates and improvements in student-staff ratios. The University Act (2002) also fixes a set of issues to be addressed in institutional plans such as strategic goals, co-operation with other universities and knowledge transfers.
Scotland (UK)	Qualitative and quantitative criteria used in performance contracts include: • equality: Admission targets for students from diverse backgrounds. Innovation: The number of research grants and contract income received; the share of UK competitive research council income; and the use of innovation vouchers for particular science to business collaborations • graduates employability: The number of first degree qualifiers; the number of undergraduate entrants in science, technology, engineering, and mathematics (STEM) courses; the development of an on-campus "employability and enterprise hub"; and the development of an employability award as part of an alumni mentoring programme.	Outcome agreements run for three years and are made between the Scottish Funding Council and individual HEIs. The outcome agreements also set annual targets for priority areas that apply to institutions. In 2014–15, there were four main priority areas: equality (opportunity), innovation, graduate employability and enterprise, as well as sustainable institutions. Universities have defined quantitative indicators for the purpose of monitoring their performance.

Source: De Boer, H., B. Jongbloed, P. Benneworth, L. Cremonini, R. Kolster, A. Kottmann, K. Lemmens-Krug, and H. Vossensteyn (2015), "Performance-based Funding and Performance Agreements in Fourteen Higher Education Systems: Report for the Ministry of Education, Culture and Science", Center for Higher Education Policy Studies, No. C15HdB014I, Enschede, Available at: http://doc.utwente.nl/93619/7/jongbloed%20ea%20performance-based-funding-and-performance-agreements-in-fourteen-higher-education-systems.pdf (Accessed: 18 October 2016).

1.4. Setting evaluation criteria, conducting evaluations and monitoring performance

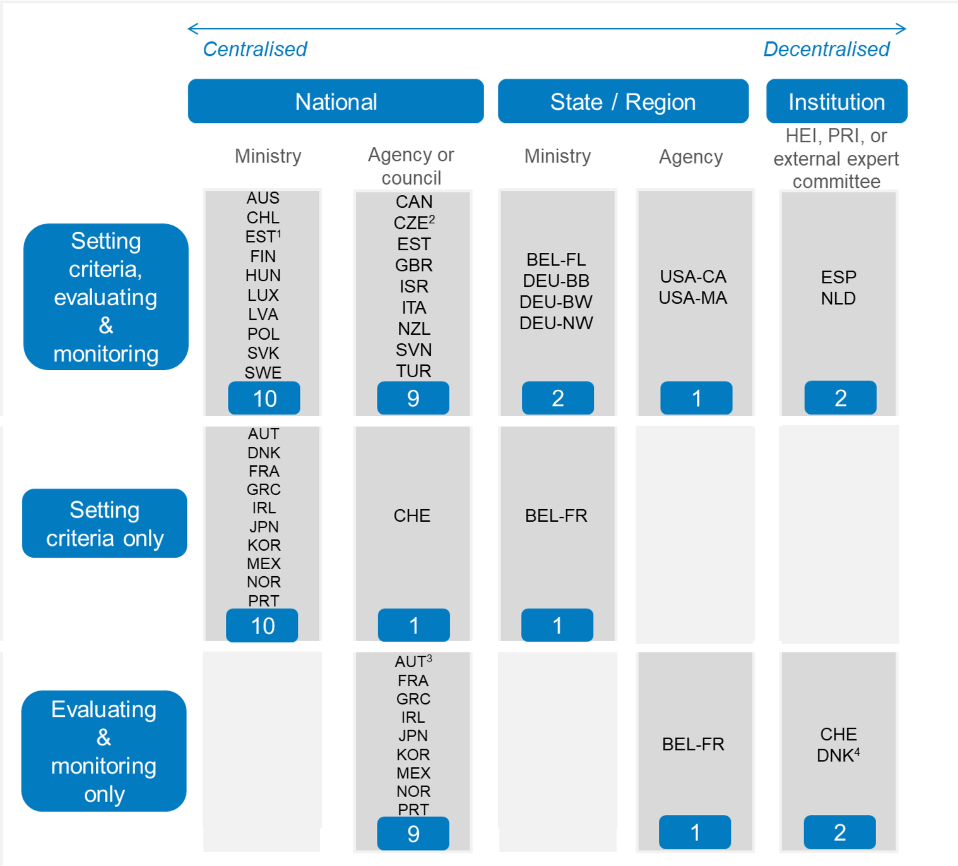
Another important component of governance of public research relates to the setting of evaluation criteria and the conduct of ex ante and ex post evaluations and monitoring of performance of HEIs and PRIs. In 23 of 34 (68%) OECD countries, national or regional Ministries of Education, Research and/or Innovation set criteria for assessing HEI performance. 10 (or 34%) of those national Ministries and Ministries in the Flemish Region in Belgium are also responsible for conducting evaluations and monitoring of performance. The other 12 countries have Ministries that set criteria while other bodies such as e.g. agencies, councils or institutions conduct evaluations. In Belgium and Germany, regional ministries set criteria for evaluations while in the United States regional agencies are responsible here (Figure 5).

Dedicated agencies are increasingly important for systematic evaluation and monitoring of performance of HEIs and PRIs. 19 of 34 countries (59%) and Wallonia in Belgium have such dedicated agencies to conduct evaluations and monitor HEIs' and PRIs' performance in place. In 10 countries (29%), the same agency also sets the criteria to be used for evaluations. Independent evaluation agencies are in place in Austria, Canada, the Czech Republic, France, Germany, Greece, Ireland, Israel, Italy, Japan, Korea, Mexico, New Zealand, Norway, Portugal, Slovenia, Switzerland, Turkey, and the United Kingdom. Austria (2012), France (2006), Ireland (2012), Italy (2010), and Portugal (2007) have recently established such agencies or replaced existing agencies with new ones.

Examples of evaluation agencies include the High Council for the Evaluation of Research and Higher Education (HCERES) in France and the Higher Education Authority (HEA) and the Quality and Qualifications Ireland (QQI) in Ireland. In Ireland, the QQI oversees quality assurance while the HEA is the agency in charge of system governance and institutional block funding for HEIs'. Both the HEA and QQI conduct quality and strategic evaluations of HEIs and PRIs based on criteria set by the State. In the Czech Republic, the Council for Research and Development, the main advisory body for the Czech government with regard to research, development and innovation (RDI) policy, is in charge of setting criteria, evaluating, and monitoring HEIs. In the Netherlands, although HEIs conduct evaluations of their performance mainly themselves, the independent Higher Education and Research Review Committee carries out evaluation of performance targets established in performance contracts that bind 7% of HEIs' budget.

Regarding monitoring, i.e. supervising the progress of HEIs/PRIs in meeting established targets, the same ministry or agency that conducts evaluation is also responsible for monitoring performance in all but one country. In 20 of 34 (59%) OECD countries, national or regional agencies monitor HEI performance, while in 12 of 34 (35%) countries national or regional ministries do so (Figure 5). In Austria, for instance, the Ministry of Science, Research, and the Economy conducts monitoring of institutions while the agency AQ Austria (Agentur für Qualitätssicherung und Akkreditierung Austria) conducts evaluations of institutions.

Figure 5. Institutions in charge of setting evaluating criteria, and evaluating and monitoring performance of HEIs



Note: This figure corresponds to questions 1.4.a to 1.4.c ("Who decides on the following key evaluation criteria of HEIs and PRIs? a) Setting criteria to use when evaluating performance of HEIs, b) Evaluating HEIs' performance, and c) Monitoring HEIs' performance"). Information is missing for Iceland.

[1] In Estonia, independent panels of scientists, e.g. foreign experts, evaluate HEIs on behalf of a Ministry.

[2] In the Czech Republic, the Council for Research and Development is responsible.

[3] In Austria, evaluations of HEIs are conducted by an independent agency, while the Ministry for Science, Research, and the Economy monitors the progress of HEIs towards achieving performance targets.

[4] In Denmark, criteria are set by the by the Ministry of Higher Education and Science in close cooperation with the institutions, while institutions themselves conduct their evaluations. The Danish Agency for Institutions and Educational Grants monitors the progress of HEIs towards achieving performance targets.

Interpretation of the figure: First row, first column shows that in 10 countries the national ministry level sets criteria to use when evaluating HEI performance but also conducts those evaluations and monitors HEIs' performance.

2. POLICY CO-ORDINATION MECHANISMS

This section describes the following institutional arrangements that are in place across OECD countries to co-ordinate policy action regarding HEIs and PRIs: i) research and innovation councils (2.1); ii) national research and innovation strategies or plans (2.2); and iii) inter-agency joint programming in research and innovation that describes arrangements between implementing agencies that result in joint action (2.3).

2.1. Research and innovation councils

Many countries have research and innovation councils, i.e. non-temporary public bodies outside of ministries and agencies that have explicit mandates to engage in one or several of the following activities: provide policy advice or oversee policy evaluation, co-ordinate policy areas relative to public research, set policy priorities and/or engage in joint policy planning regarding HEI and PRI policies. 31 of 35 OECD countries (or 89%) have such institutions. 15 countries have created their current councils after 2007, a reflection of the importance these bodies are given by policy (Table 2). Germany has several major councils that advice on matters relevant to research.[4] Aside from its Federal Science Policy Council, Belgium also has three regional councils. In most countries, the current council replaced an older council of different characteristics. In Demark, for instance, the Council for Research and Innovation Policy that was created in 2014 replaced the Danish Council for Research Policy of 2003. In Finland, the Research and Innovation Council of 1987 was dissolved in 2016 and replaced by a new Research and Innovation Council with different mandate and composition in the same year.

Councils differ in their mandate. In 28 of the 31 countries (90%) with councils, the councils provide policy advice and for 6 of them, those of Australia, Canada, Denmark, the Netherlands, Portugal and Sweden, this is their core role. In 23 of 31 countries with councils (or 74%), the council also develops strategic priorities. 15 of 31 (48%) councils, the councils conducts evaluation of policy reforms. In 15 of 31 (48%) countries, the council is also tasked with policy co-ordination between within government and with non-state stakeholders (Figure 6). The councils in Czech Republic, Israel, Japan, Korea, Mexico and Turkey have broader mandates than others as they also decide how budgets are allocated.

Ireland, Italy, New Zealand and Norway do not have research and innovation councils but use other forms of horizontal co-ordination. In the case of Norway 8 temporary sectoral strategy committees for STI called “21 Forums” were created. The committees include representatives from the private sector, academia and the government. They provide policy advice, formulate strategies that they submit to the government and serve as stakeholders forums for policy consultation. Since 2014, Norway also holds annual summits on specific STI policy themes. The summits gather stakeholders from the private sector, academia, civil society and the public sector and aim at stimulating the policy debate and co-ordination on topics related to the national long-term plan for research and higher education. They are chaired by the Prime Minister and ministers participate.

Table 2. Main research and innovation councils and their year of creation

Country	Main research and innovation councils	Year of creation
Australia	Commonwealth Science Council	2014
Austria	Austrian Council for Research and Technology Development	2000
Belgium	Federal Council for Science Policy	1997
Belgium - Brussels-Capital Region	Council of Science Policy	n.a.
Belgium - Flemish Community	Flemish Council for Science and Innovation	n.a.
Belgium – Walloon Region	Science Policy Pole	n.a.
Canada	Technology and Innovation Council	2007
Chile	National Innovation Council for Development	2005
Czech Republic	Council for Research, Development and Innovation	2009
Denmark	Council for Research and Innovation Policy	2014
Estonia	National Science and Technology Council	2002
Finland[1]	Research and Innovation Council (dissolved)	1987-2016
	Research and Innovation Council (new)	2016
France	Strategic Research Council	2013
Germany	Innovation Dialogue	2010
	Council of Science and Humanities	1957
	Expert Commission for Research and Innovation	2007
Greece	National Council for Research and Innovation	1984
Hungary	Research Council, Innovation Council	2015
Iceland	Science and Technology Policy Council	2003
Israel	Higher Education Council	1958
Japan	Council for Science, Technology and Innovation	2001
Korea	National Science and Technology Council	1973
Latvia	Latvian Science and Innovation Strategic Council	2014
Luxembourg [2]	Superior Committee for Research and Innovation	2008-14
Mexico	General Council for Scientific Research, Technological Development and Innovation	2002
Netherlands	Advisory Council for Science, Technology and Innovation	n.a.
Poland	Council for Innovation	2016
Portugal[3]	National Council for Science and Technology, National Council for Entrepreneurship and Innovation	2012
Slovak Republic	Government Council of the Slovak Republic for Science, Technology and Innovation	2011
Slovenia	National Science and Technology Council	2002
Spain	Council for Scientific, Technological and Innovation Policies	2012
Sweden	National Council for Innovation and Quality in the Public Sector	2015
Switzerland	Swiss Science Council	1965
Turkey	Supreme Council for Science and Technology	1983
United Kingdom	The British Council for Science and Technology	2005
United States	President's Council of Advisors on Science and Technology	1990

Note: Ireland, Italy, New Zealand and Norway have no research and innovation council as defined in this report. Information is not available on the year of establishment of councils in the Belgium-Capital Region, the Flemish Community in Belgium, and the Walloon Region in Belgium. Information for all three German Councils is used for the cross-country comparison.

[1] The Finnish Research and Innovation Council had been dissolved in 2016 and a new Council was established under the same name in the same year. Due to changes to the composition and mandate of the Council this analysis treats them as two separate entities.

[2] In Luxembourg, the Superior Committee for Research and Innovation is inactive since 2014.

[3] In Portugal, the National Council for Science and Technology and the National Council for Entrepreneurship and Innovation are inactive since 2015.

Figure 6. Mandates of the research and innovation councils

	CZE	HUN	JPN	MEX	DEU[1]			GBR	ISR	KOR	LVA	POL	TUR	BEL[2]				CHL	FIN[3]		LUX[4]	SVN	SVK	AUT	CHE	ESP[5]	EST	FRA	GRC	ISL	USA	AUS	CAN	DNK	NLD	PRT[6]		SWE	Share of countries with Councils (%)
					Innovation Dialogue	Council of Science and Humanities	Expert Commission for Research and Innovation							BEL-FED	BEL-BC	BEL-FL	BEL-WA		Old council (86'-16')	New council (since 16')																National Council for Science and Technology	National Council for Entrepreneurship and Innovation		
Policy advice	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X		X		X		X	X	X	X	X	X	X	X	90% 28 of 31
Strategic priority setting	X	X	X	X		X	X	X	X	X	X	X	X		X	X	X	X	X		X	X	X			X	X	X	X	X	X								74% 23 of 31
Policy evaluation	X	X	X	X		X	X	X	X	X	X	X			X	X	X				X	X		X	X														48% 15 of 31
Policy coordination	X	X	X	X	X	X	X	X			X	X	X					X	X				X			X		X		X									48% 15 of 31
Decision on budget allocations	X	X	X	X					X	X			X																										23% 7 of 31

Note: This figure corresponds to questions 2.2.a – 2.2.e ("Does the Research and Innovation Council's mandate explicitly include: a) Policy coordination, b) Preparation of strategic priorities, c) Decision-making on budgetary allocations, d) Evaluation of policies' implementation (including their enforcement), e) Provision of policy advice?"). There is no Council in Ireland, Italy, Norway and New Zealand. Percentages are expressed as a share of the countries with a council in place (*N*=31).

[1] Germany has three main Councils: The Council of Science and Humanities, the Expert Commission for Research and Innovation and the Innovation Dialogue. Information for all three councils was used for the cross-country comparison. All three councils' mandates include policy coordination and provision of policy advice. The mandates of the Council of Science and Humanities and the Expert Commission for Research and Innovation also include development of strategic priorities and policy evaluation.

[2] Belgium has a Council for the Brussels-Capital Region (Council of Science Policy), a Council for the Flemish Community (Flemish Council for Science and Innovation) and a Council for the Walloon Region (Science Policy Pole) and a Federal Council (Federal Science Policy Council).

[3] The Finnish Research and Innovation Council had been dissolved in 2016 and a new Council was established under the same name in the same year. Due to changes to the composition and mandate of the Council this analysis treats them as two separate entities.

[4] In Luxembourg, the Superior Committee for Research and Innovation has not convened since 2014.

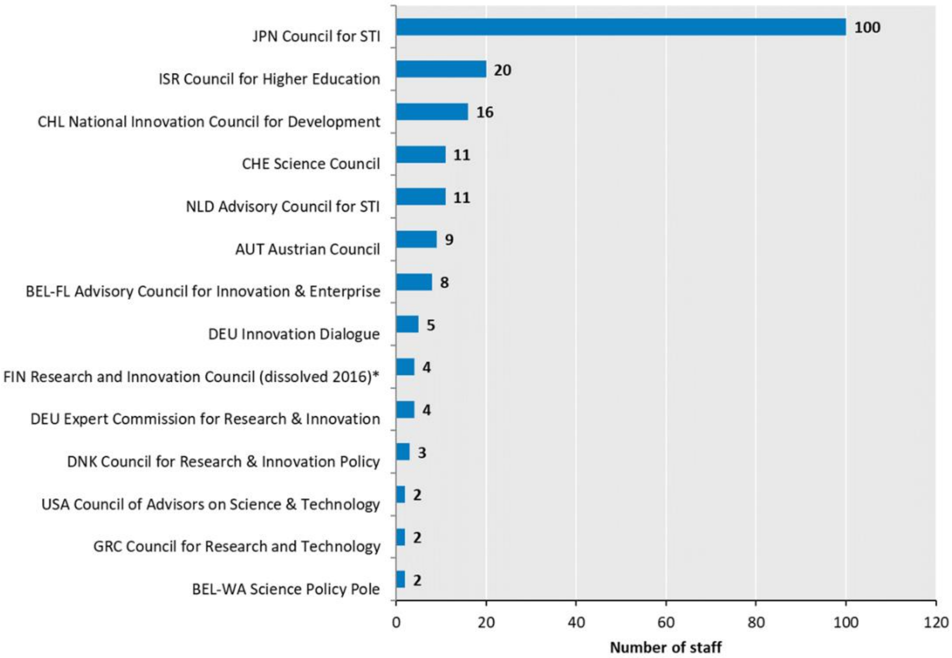
[5] Spain's Council for Scientific, Technological and Innovation Policies is responsible for the coordination of national and regional science, technology and innovation policies. Policy Advice is provided by a dedicated Advisory Council.

[6] Portugal has two main councils: The National Council for Science and Technology, and the National Council for Entrepreneurship and Innovation. They have not convened since 2015.

Interpretation of the figure: First row shows that in all countries except for France, Iceland, and Spain the research and innovation council has a mandate for policy advice, among others.

Councils also differ with regard to their staff and budgets. In 14 of 30 (or 47%) countries for which information is available, councils have own staff (i.e. secretariat). Of these councils, aside from Japan all have 20 or less staff members and 15 have no direct staff (Figure 7). 10 out of 29 councils (or 34%) for which information is available and the Councils of the Brussels-Capital Region and the Flemish Region in Belgium have their own budget. The size of the annual budget available differs. The councils in Austria, Chile, Israel, Japan and the Netherlands have annual budgets of USD 1.3 million (EUR 1 million) or more, while the councils in Belgium-Brussels, Belgium-Flanders, Denmark, the German Expert Commission for Research and Innovation have budgets of below USD 1.3 million (EUR 1 million) per year.

Figure 7. Number of staff of research and innovation councils



Councils without staff for which information is available:

AUS	Commonwealth Science Council	KOR	National Science & Technology Council
BEL	Federal Council for Science Policy	LVA	Science & Innovation Strategic Council
BEL-BC	Council of Science Policy	MEX	General Council
CZE	Council for Research, Development & Innovation	POL	Council for Innovation
EST	National Science and Technology Council	PRT	National Council for Science & Technology, National Council for Entrepreneurship & Innovation
ESP	Council for Science, Technology and Innovation Policies	SVK	Government Council for Science, Technology, and Innovation
FIN	Research & Innovation Council (since 2017)*	SVN	Council for Higher Education
FRA	Strategic Research Council	SWE	Council for Innovation & Quality in the Public Sector
HUN	Research Council and Innovation Council under the National Research, Development & Innovation Office	TUR	Supreme Council for Science & Technology

Note: The figure corresponds to question 2.4.a ("Does the Council have its own staff? If so, please indicate the number of staff") and showcases only countries where the Council has its own staff. The Council has its own staff in Canada but no information on the number of staff is available. Information on staff numbers is not available for Canada, Iceland, Luxembourg, and the United Kingdom. Information on budget numbers is not available for Belgium, Finland, Iceland, Luxembourg, and the United States. In Luxembourg, the Superior Committee for Research and Innovation is inactive since 2014. There is no Council in Ireland, Italy, New Zealand, and Norway.

* The Finnish Research and Innovation Council had been dissolved in 2016 and a new Council was established under the same name in the same year. Due to changes to the composition and mandate of the Council this analysis treats them as two separate entities.

2.2. STI plans or strategies

An alternative mechanism for co-ordination across different stakeholders for research policy is STI plans or strategies that outline a common strategy. Most countries (33 of 35, 94%) have national STI strategies or plans (that have larger coverage than industry and research specific strategies.[5] Of these, 22 are EU member countries or participants in Horizon 2020 and the European Commission's Framework Programmes for whom having such plans is a requirement (see discussion below). Most countries also have additional STI strategies in place (33 of 35, 94%), often with a more specific regional, sectoral, thematic or transnational focus (e.g. participation in international research infrastructure projects, or increase in the number of European Research Council grants received). Hungary, for instance, has a national STI strategy and also a Research and Innovation Strategy for Smart Specialisation. Scientific and thematic priorities are included in the latter strategy. The EU's strategic agenda also have an important impact on EU countries and participants (Box 2).

In most countries, STI strategies address major societal challenges including demographic change, health, environment, and smart transport and cities (30 of 33 countries or 91% and the Belgian Brussels-Capital Region, the Flemish Region and the Walloon Region). 25 of 33 countries or 76%, the Brussels-Capital Region and the Walloon Region in Belgium stress in their STI strategy the need for research and innovation to develop a sustainable economy. 13 of 33 countries (or 40%) emphasise in their strategy the importance of STI to help address demographic change. The STI strategies of 15 of 33 countries (or 45%) also refer to investing in STI to help improve transport systems.

Strategies also generally identify specific scientific research, technologies or economic fields (31 of 33 countries or 94% and the Belgian Brussels-Capital Region, the Flemish Region and the Walloon Region), mentioning energy and energy technologies, health and life sciences, information and communication technologies (ICT), nanotechnology and advanced materials. In 29 of 33 countries (or 88%), in the Belgian Brussels-Capital Region, and in the Flemish Region in Belgium, national plans focus on the digital economy (Guellec and Paunov, 2018).

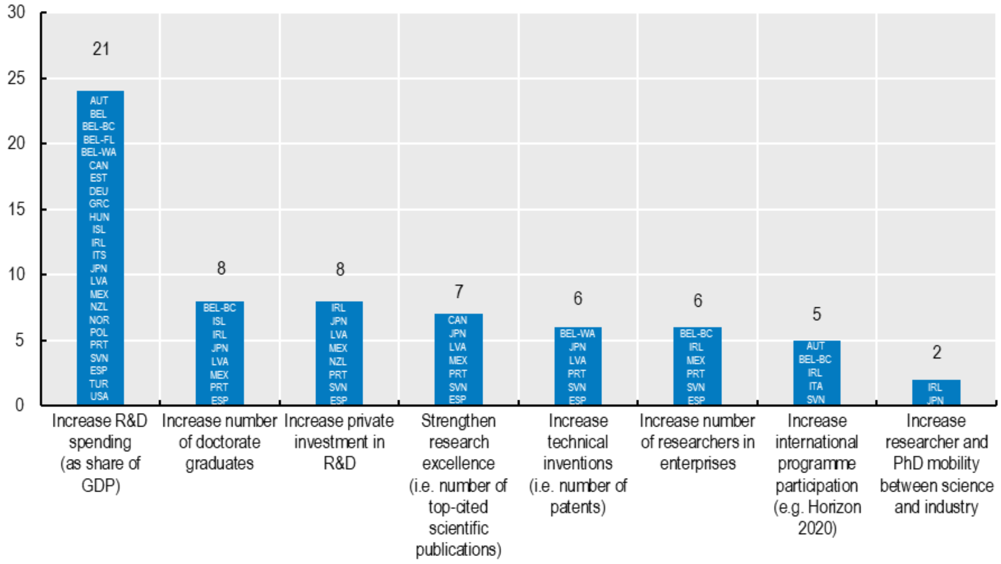
In 23 of 32 countries (72%) that provided such information, STI strategies also address priorities for specific Federal States or regions, i.e. regional Research and Innovation Strategy for Smart Specialisation (RIS). To develop a RIS is a prerequisite in order to receive funding from the European Regional Development Fund (ERDF), which constitutes an important funding source from the European Union to European regions. These local STI strategies identify diverse scientific, research, technology and innovation priorities across regions. The Smart Specialisation strategy of the Walloon Region in Belgium, for instance, addresses STI priorities for aeronautics, green technologies, health, mechanical engineering, as well as transport and logistics, while the Norte Region in Portugal focuses on STI in automotive, cultural and creative industries as well as the ICT sector. Smart Specialisation strategies have also been taken up at country level in Estonia, Hungary, Luxembourg and Slovenia.

Some non-EU member regions also participate in the Framework Programme and have RIS strategies, notably Norway and Turkey. Dedicated strategies for regional development are also in place in Australia, Canada, Japan, and Mexico. For example, Mexico has adopted three Regional Agendas for Innovation and 32 Agendas for Innovation at the level of Federal States since 2014. These regional strategies in Mexico were developed following the EU's RIS methodology.

National STI strategies also include quantifiable targets that make it possible to assess whether policy has met them or not. In 21 of 33 countries (64%) and in the Brussels-Capital Region, the Flemish Region and the Walloon Region in Belgium, national STI strategies include such quantitative targets. National STI strategies of EU Member countries refer explicitly to the Horizon 2020 objective, which is to raise expenditures on R&D to 3% of GDP. Note that the European Council has recommended that EU Member States set their own targets for R&D investment. 21 out of 33 countries (or 64%) with national STI strategies and the Brussels-Capital Region, the Flemish Region and the Walloon Region in Belgium list the R&D spending target.

The data also shows that 12 (or 36%) of national STI strategies and the STI strategy of Brussels-Capital Region in Belgium include targets for HEIs and PRIs. These include raising funding for doctoral students (7 of 33 and the Brussels-Capital Region in Belgium), and job placements for researchers and PhDs in industry (5 of 33 and the Brussels-Capital Region in Belgium) (Figure 8). In Japan and Ireland, the national STI strategy sets quantitative benchmarks for knowledge transfer between universities and industry including raising the private funding of university research, the amount of collaborative research funds received from industry, and the number of license agreements on university patents. Japan's 5th Science and Technology Basic Plan for 2016-20 includes targets to increase the number of tenure positions for young researchers at universities and to raise the share of female researchers among new hired university personnel. In Iceland, the main STI strategy, the "Science and Technology Policy and Action Plan 2014-16" includes plans to increase of competitive funding as a share of funding of HEIs and PRIs to one third of total funding.

Figure 8. Quantitative targets included in national STI strategies



Note: The figure corresponds to question 2.6.e ("Does the national STI strategy or plan address any of the following priorities? Quantitative targets for monitoring and evaluation (e.g. setting as targets a certain level of R&D spending for public research etc.)?) and showcases only countries where the national STI strategies have quantitative targets. Information is missing for Australia, the Slovak Republic, and the United Kingdom. Israel and Luxembourg do not have national STI strategies. Flanders also targets an increase R&D spending (as a share of GDP).

Box 2. The EU strategies for growth and innovation

Europe 2020 is EU's strategic agenda for smart, sustainable and inclusive growth and job creation. Europe 2020 sets five quantitative objectives; one of them refers to STI explicitly: setting as objective the need to raise investment in R&D to 3% of GDP by 2020. Specific national targets that reflect these goals were adopted by all EU member countries via their respective National Reform Programmes (NRP). These goals are also reflected in many EU countries' STI strategies.

Regarding innovation, the Innovation Union is a flagship initiative of the Europe 2020 Strategy. Horizon 2020 is one of the main funding programming of Europe 2020. Horizon 2020 aims to enhance the competitiveness of Europe's research, remove barriers to innovation, facilitate public-private partnerships for innovation and address social challenges. It contributes to the integration of the European Research area (ERA).

The non-EU member countries, Israel, Norway and Turkey also participate in Horizon 2020. Norway has included Horizon 2020 objectives in its national STI strategic plan. Norway also has a bilateral arrangement for participation in interregional programmes under the EU's Regional Policy. Israel participates in the Horizon 2020 programme and signed the agreement with the EU in 2014. Turkey is associated with the European Commission (EC) Framework Programme since 2003 and the Horizon 2020 programme since 2014.

Source: European Commission (2016a), "European Structural & Investment Funds", http://ec.europa.eu/contracts_grants/funds_en.htm (accessed on 16 November 2016); European commission (2016b), "Horizon 2020 – the EU Framework Programme for Research and Innovation", https://ec.europa.eu/programmes/horizon2020/en/ (accessed on 16 November 2016); European Commission (2016c), "Registered countries and regions in the S3 Platform", http://s3platform.jrc.ec.europa.eu/s3-platform-registered-regions (accessed on 16 November 2016); European Commission (2015a), "EU 2020 Targets", http://ec.europa.eu/europe2020/europe-2020-in-a-nutshell/targets/index_en.htm (accessed on 16 November 2016); European Commission (2014), Turkey joins Horizon 2020 research and innovation programme, http://europa.eu/rapid/press-release_IP-14-631_en.htm (accessed on 5 March 2017). European Commission (2016e), "Smart Specialisation", https://ec.europa.eu/jrc/en/research-topic/smart-specialisation (accessed 16 November 2016); The Norwegian Mission to the EU (2016), "Norway's participation in EU programmes and agencies", www.eu-norway.org/eu/Coopperation-in-programmes-and-agencies/#.WCxKrMlUsXh (accessed on 16 November 2016). Delegation of the European Union to Israel (N.D.), "Scientific cooperation", http://eeas.europa.eu/archives/delegations/israel/eu_israel/scientific_cooperation/index_en.htm (accessed on 16 November 2016).

Table 3. Names and year of establishment of main national STI strategies

Country	Main national STI strategies	Period
Australia	National Innovation and Science Agenda	since 2015
Austria	Austrian National RTI Strategy "Becoming an Innovation Leader"	2011-20
Belgium	Federal Government Agreement	since 2014
Belgium - Brussels-Capital Region	Regional Innovation Plan	2016-20
Belgium - Flemish Community	Flanders Reform Programme EU 2020, Flanders Pact 2020	2008-20
Belgium – Walloon Region	Wallonia Marshal Plan 4.0 2015-19	2015-19
Canada	Seizing Canada's Moment: Moving Forward in Science, Technology and Innovation (2014), Canada’s Innovation and Skills Plan (2017)	2014-25
Chile	Science, Technology and Innovation for a New Pact of Sustainable and Inclusive Development	2017-30
Czech Republic	International Competitiveness Strategy 2012-20	2012-20
Denmark	Denmark – Ready to seize future opportunities: The Government’s objectives for Danish research and innovation	since 2018
Estonia	The Estonian Research and Development and Innovation Strategy “Knowledge-based Estonia” 2014-20	2014-20
Finland	Research and Innovation Policy Guidelines 2015-20 (2014), Research and Innovation Council’s Vision and Roadmap for 2030 (2017)	2014/17-20/30
France	National Research Strategy	2015-20
Germany	High-Tech Strategy	since 2014 (updated)
Greece	National Research and Innovation Strategy For Smart Specialization 2014-20	2015-20
Hungary	National Research, Development and Innovation Strategy 2013-20 - Investment in the Future	2013-20
Iceland	Research and Development Policy and Action Plan 2017-19	2017-19
Ireland	Innovation 2020: Excellence Talent Impact	2015-20
Italy	National Research Plan (NRP) for 2014-20; Horizon 2020 Italia (2014-20); and the National Smart Specialization Strategy (2014-20)	2014-20
Japan	The Fifth Science and Technology Basic Plan 2016-20	2016-20
Korea	The Fourth Science and Technology Basic Plan 2018-22	2017-22
Latvia	Guidelines for the Development of Science, Technology and Innovations 2014-20.	2014-20
Mexico	The Special Programme for Science, Technology and Innovation 2014-18	2014-18
Netherlands	Enterprise Policy “To the Top” (2013), National Research Vision (2014), “Working Together for Renewal” (2015), Strategic Agenda for Higher Education, Research and Science (2015), and National Research Agenda (2015)	2013/15-20/25
New Zealand	National Statement of Science Investment (NSSI)	2015-25
Norway	Long -term plan for research and higher education 2015–24	2014-24
Poland	National Reform Programme (2011), National Research Programme (2011), Strategy for Responsible Development (2017)	2017-20
Portugal	National Research and Innovation Strategy for Smart Specialisation, National Plan for Science, Technology and Innovation 2017-20	2018-30
Slovak Republic	Research and Innovation Strategy for Smart Specialisation 2014-20	2014-20
Slovenia	Research and Innovation Strategy of Slovenia 2011-20 (2011), Slovenia’s Smart Specialisation Strategy (2015), Digital Slovenia 2020 (2016)	2011-20
Spain	Spanish Strategy for Science, Technology and Innovation 2013-20	2013-20
Sweden	National Innovation Strategy (2012); Government Bill Research and Innovation (2016)	2016-20
Switzerland	Education, Research and Innovation Policy Dispatch of the Federal Council 2017-20	2016-20
Turkey	National Science, Technology and Innovation Strategy 2011-16	2011-16
United Kingdom	Our Plan for Growth: Science and Innovation	since 2014
United States	Driving towards Sustainable Growth and Quality Jobs	since 2015 (updated)

2.3. Inter-agency programming

Inter-agency programming, i.e. formal arrangements between agencies that result in joint action, is a new policy instruments to co-ordinate policy action at the level of implementing agencies. Examples are funding programmes between funding agencies. In Sweden, VINNOVA, the Swedish Energy Agency and the Swedish Research Council Formas jointly implemented the Swedish Strategic Innovation Area (SIO) initiative in 2012 (OECD, 2016b). Under the initiative, the three agencies jointly selected and funded research projects that have to be submitted by university-industry consortia. In the United States, the Interagency Working Group on the Science of Science Policy (SoSP) coordinates evaluation and impact assessment efforts of several agencies (Box 3).

Box 3. Co-ordination of science policy between US government agencies and the academic community

The Interagency Working Group on the Science of Science Policy (SoSP) is responsible for setting-up evaluation standards and joint data infrastructure for impact assessments of publicly-funded research across U.S. government departments and agencies. The SoSP's activities include:

- Developing jointly tools, theories, methodologies, and collect and exchange data to evaluate projects where federal agencies can collaborate and join efforts.
- Developing a federal STI policy information infrastructure (STAR METRICS) to enable the SoSP community to assess the impacts of STI programmes and federal policies.
- Developing a central, government-wide profile of federal government researchers and for other researchers funded by science agencies to document, publicise, and measure the attainment of federal goals associated with STI activities.
- Conducting workshops with recipients of grants from the National Science Foundation (NSF)'s Science of Science and Innovation Policy (SciSIP) programme.

Participating institutions of the SoSP include the Departments of Agriculture, Commerce, Defence, Education, Energy, Health and Human Services, Homeland Security, Interior, Transportation, Veterans Affairs, the State Department, the Environmental Protection Agency, the National Aeronautics and Space Administration (NASA), and the National Science Foundation (NSF). Organisations in the Executive Office of the President are also represented in the Working Group (Office of Management and Budget, and the Office of Science and Technology Policy). SoSP is co-chaired by the Department of Energy and the NSF.

Source: OECD (2016a); SoSP (2016).

3. EXTERNAL STAKEHOLDERS INVOLVEMENT AND INSTITUTIONAL AUTONOMY

This section describes how the private sector, civil society and HEIs and PRIs are involved in the public-research policy-making process in their representation in councils and university boards (3.1), and discusses online consultation platforms as new mechanisms to engage stakeholders in policy making (3.2). It also discusses the institutional autonomy of HEIs and PRIs (3.3) which critically determines HEIs/PRIs own role in policy design and implementation.

3.1. Participation in councils and university boards

Participation in the research and innovation councils shapes the involvement of external stakeholders – defined as stakeholders outside of government - in policy design. 29 of 31 (94%) of countries involve HEIs and/or their associations, while 24 (77%) involve PRIs and/or their associations. Private sector representation in councils is widespread across the OECD and more common than civil society representation: 26 countries (or 84%) include private sector representatives in their main councils. Representatives from civil society and citizen associations (NGOs, unions) are involved in 15 of 31 (48%) countries. (Figure 9).

Foreign experts only participate in 6 of 31 (19%) OECD countries with councils (Austria, France, Germany, Greece, Switzerland and the United Kingdom). In most cases these experts are academics but this is not always the case. The Austrian Council, for instance, includes a British entrepreneur and a representative from the Swedish Government Agency for Innovation Systems (VINNOVA).

With regards to government representation, 16 of 31 (84%) countries have government representatives in at least one of their councils. The head of state or prime minister are council members in 13 of 31 (39%) of countries, and ministers in 21 of 31 (68%) of countries.

A number of countries have councils with non-governmental representatives only, notably the Czech Republic, Denmark, Hungary, the Netherlands, Switzerland and the United States. This includes HEI and PRI representatives, private sector representatives and/or civil society organisations.

Federal State/regional governments are represented in 8 of 31 countries (26%, i.e. Belgium, Canada, France, Germany, Greece, Latvia, Spain and the United Kingdom) and 5 of those are countries with federal structure. In these country contexts, the councils play a role in policy co-ordination regarding HEIs and PRIs between national government and subnational authorities. Spain's Council, the Council for Scientific, Technological and Innovation Policies (CPCTI), for instance, is responsible for policy coordination between the national government and the Autonomous Communities. In the Spanish case, policy advice is provided by a specific Advisory Council, which involves representatives of main stakeholders from science and industry.

Figure 9. Who formally participates in the research and innovation council?

		SWE	LVA	FIN[1] Old council (86'-16')	FIN[1] New council (since 16')	FRA	JPN	EST	KOR	AUS	ISL	MEX	SVK	TUR	DEU[2] Innovation Dialogue	DEU[2] Council of Science and Humanities	DEU[2] Expert Commission for Research and Innovation	AUT	SVN	LUX[3]	PRT[4] National Council for ST	PRT[4] National Council for Entrepreneurship and Innovation	CHL	ESP	ISR	POL	GRC	BEL[5] BEL-FED	BEL[5] BEL-BC	BEL[5] BEL-FL	BEL[5] BEL-WA	CAN	GBR	HUN	CZE	CHE	NLD	DNK	USA	Share of countries with Councils (%)
Government representatives	Head of State/Prime Minister	X	X	X	X	X	X	X		X	X	X	X	X	X																									39% 12 of 31
	Ministers	X	X	X	X	X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X	X	X	X														68% 21 of 31
	Other national government officials		X	X	X	X		X	X				X	X		X						X	X	X					X		X	X	X							45% 14 of 31
	Funding agencies representatives			X	X				X			X	X	X								X	X									X	X							29% 9 of 31
	Local and regional government		X			X										X								X			X				X	X	X							26% 8 of 31
Stakeholders	HEI representatives	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X		X	X	X		X	X	X	X	X	X	X	X	X	94% 29 of 31
	PRI representatives		X	X	X	X	X		X	X		X	X	X	X	X	X	X	X	X	X	X					X	X	X		X	X	X	X	X	X	X	X	X	77% 24 of 31
	Private sectors representatives	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X				X	X	X		X	X	X	X			X	X	X	84% 26 of 31
	Civil society	X		X		X					X		X		X	X			X	X			X					X	X		X	X	X	X				X	X	48% 15 of 31
	Foreign experts					X											X	X									X						X			X				19% 6 of 31

High-level policy representation (Head of State/Prime Minister, Ministers)

Stakeholders only

Note: This figure corresponds to question 2.3. ("Who formally participates in the Research and Innovation Council?"). There is no research and innovation council in Ireland, Italy, Norway and New Zealand. Percentages are expressed as a share of countries with a council in place (N=31).

[1] The Finnish Research and Innovation Council had been dissolved in 2016 and a new Council was established under the same name in the same year. Due to changes to the composition and mandate of the Council this analysis treats them as two separate entities.

[2] Germany has three main Councils: The Council of Science and Humanities, the Expert Commission for Research and Innovation and the Innovation Dialogue. Information of all three councils was used for the cross-country comparison. All three councils' mandates include policy coordination and provision of policy advice. The mandates of the Council of Science and Humanities and the Expert Commission for Research and Innovation also include development of strategic priorities and policy evaluation.

[3] In Luxembourg, the Superior Committee for Research and Innovation has not convened since 2014.

[4] Portugal has two main councils: The National Council for Science and Technology, and the National Council for Entrepreneurship and Innovation. They have not convened since 2015.

[5] Belgium has a federal Council (Federal Science Policy Council) Council for the Brussels-Capital Region (Council of Science Policy), a Council for the Flemish Community (Flemish Council for Science and Innovation) and a Council for the Walloon Region (Science Policy Pole).The Federal Science Policy council is composed of experts participating in their own capacity but they are issued from the academia, the private sector or policy circles.

Interpretation of the figure: First row shows that in Sweden, Latvia, Finland, France, Japan, Estonia, Australia, Iceland, Mexico, Slovak Republic, Turkey, and in Germany the Research and Innovation Council includes the head of state (i.e. president or prime minister), among others.

Stakeholder involvement at the HEI level is also an important element for stakeholder consultation, particularly with increased autonomy of HEIs. In most OECD countries, universities' governance structure is made of a board or senate (referred to as board here). The university board is the main decision-making body and it is in charge of priority setting. The private sector participates in governing boards of HEIs in 25 of 34 countries (74%). In most cases, business representatives are from large firms, but some countries such as Iceland and Ireland also include representatives from small and medium sized enterprises (SMEs). 23 of 34 (68%) have civil society – i.e. citizens, NGOs and foundations –

represented in university councils. 32% of countries (11 of 34) have both private sector and civil society representatives in university councils. 10 countries have foreign experts in their university boards (29%). In 3 countries (or 9%), only the private sector is represented. In Italy and Korea, only civil society is present. 6 countries (18%) do not have external stakeholders represented (Figure 10).

External stakeholder representation has started recently in some countries. For instance, in Portugal university reforms of 2007 introduced their representation in the governing boards of HEIs. In France, the representation of business, labour unions and local actors in governing bodies of HEIs and PRIs was introduced by the Law on Higher Education and Research in 2013. In Switzerland, university councils are the main executive body of HEIs and take decisions on strategic issues informing their thematic and scientific priorities. They only include external members: representatives from industry, HEIs, PRIs, and civil society.

Figure 10. Who formally participates in public university boards?

	AUS	CHE	GBR	IRL	ISR	NZL	USA		DNK	AUT	BEL-FL	CAN	ESP	FIN	ISL	NLD	NOR	PRT	SWE	POL	DEU			FRA	HUN	JPN	SVN	SVK	ITA	KOR	GRC	CHL	CZE	LUX	LVA	MEX	TUR	Share of countries with boards
							USA-MA	USA-CA													DEU-BW	DEU-NW	DEU-BB															
Private sector	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X										74% 25 of 34
Civil society	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					X	X								68% 23 of 34
Foreign experts	X	X	X	X	X	X	X	X	X																			X			X							29% 10 of 34
No formal representation																																X	X	X	X	X	X	18% 6 of 34

Private sector and civil society: 59%

No external stakeholder representation: 18%

Note: This figure corresponds to question 3.1.b ("Do stakeholders participate as formal members in governing boards of HEIs?") Information is missing for Estonia. There is no formal stakeholder participation in HEI boards in Chile, Czech Republic, Luxembourg, Latvia, Mexico, and Turkey. Information on participation in university boards is missing for Estonia. Percentages are expressed as a share of countries with information on the composition of HEI boards (N=34).

Interpretation of the figure: The first row shows that in all countries except Italy, Korea, Greece, Chile, Czech Republic, Luxembourg, Latvia, Mexico, and Turkey the private sector is represented in boards of HEIs.

3.2. Online public consultations and temporary stakeholder consultation processes

Online public consultations are a new policy instruments used to include civil society more in policy formulation processes. Online platforms were used to develop national STI plans in Australia (2016 National Research Infrastructure Roadmap), France (National Research Strategy, 2015), Japan (Fifth Science and Technology Basic Plan 2016-20), Mexico (National Development Plan 2013-18) and the Netherlands (2015 consultation for the Dutch National Research Agenda, Box 4). In 2016, the United Kingdom's Department for Business, Energy and Industrial Strategy issued an online consultation in preparation for the National Innovation Strategy. In 2017, Finland introduced online consultation to develop a national vision and roadmap "Vision for Higher Education and Research in 2030". In Denmark, 100 stakeholders from business, academia, civil society including trade unions and non-profit organisations, municipalities, and regional authorities contributed with 476 inputs on new research priorities for the RESEARCH2025 catalogue that provides a collection of political priorities for strategic research in the coming years. In Greece, several online consultation platforms were established to identify thematic priorities for the Regional Smart Specialisation Strategy more aimed at getting inputs from HEIs, PRIs and industry than the public at large.

In the United Kingdom several online consultation platforms are in place whereby ministries and regional governments issue regular calls on specific policy matters, including those targeting HEIs and PRIs. The online consultation platforms are in general open to the public. Examples of such platforms include the Consultation Hub of the Department for Business, Energy and Industrial Strategy on which the innovation strategy consultation was held in 2016, along with other consultations on themes related to education and research (e.g. a consultation on the development of a "Teaching Excellence Framework") (Department for Business, Energy and Industrial Strategy Consultation Hub, 2017). The Department for Education also has an online consultation platform in place (Department for Education Consultation Hub, 2016). The Scottish Government's online consultation platform issues questions on various policy matters some of which regard HEIs such as e.g. college governance (Scottish Government Consultation Hub, 2017).

Other more traditional forms of involving stakeholders that continue to be important include working groups, roundtables and calls for inputs. For example, Turkey introduced committees for coordination of STI programme design in 2010 (High-Level Prioritization Meetings and Focus Groups). These committees were introduced by the funding agency Scientific and Technological Research Council of Turkey (TÜBİTAK). They aim to facilitate a bottom-up process of identification of sectoral priorities and include the private sector and academia and members from the research and innovation council (the Supreme Council for Science and Technology, SCST). TÜBITAK also carried out an open ended survey to identify priorities in the biomedical technology sector, gathering upwards of 1 200 ideas from 300 researchers and experts. On the basis of these inputs technology roadmaps and policy programme were developed (OECD, 2016a).

Similarly, stakeholders participated in the preparation of Estonia's "Research and Development and Innovation Strategy 2014-20 - Knowledge-based Estonia". A Strategy Preparation Committee was formed by the Ministry of Education and Research that gathered representatives from the private sector, universities, research institutions, and state authorities. Additionally, over 200 specialists from the research, business (including entrepreneurs) and state sectors were involved in the preparation of the strategy.

Also, in the Czech Republic, for instance, private sector representatives were consulted via working groups to identify national priority fields for research. The working groups provided inputs to the main national council (i.e. Czech Council for Research and Development) but also supported work on national and regional Smart Specialisation Strategy.

Like online consultations, these temporary stakeholder involvement methods allow for broader consultation and for sectoral targeting. They complement more permanent consultations such as participation in research councils.

3.3. HEIs' and PRIs' autonomy

This section uses data on the autonomy of universities from the 2015 University Autonomy in Europe Scorecard (Estermann et al., 2015). The data covers 21 EU Member countries, Iceland, Norway, Switzerland, and Turkey. To complement, additional data where gathered for PRIs' autonomy for all OECD countries and for HEIs' autonomy for Australia, Canada, Chile, Israel, Japan, Korea, Mexico, New Zealand, and the United States to cover all OECD countries.

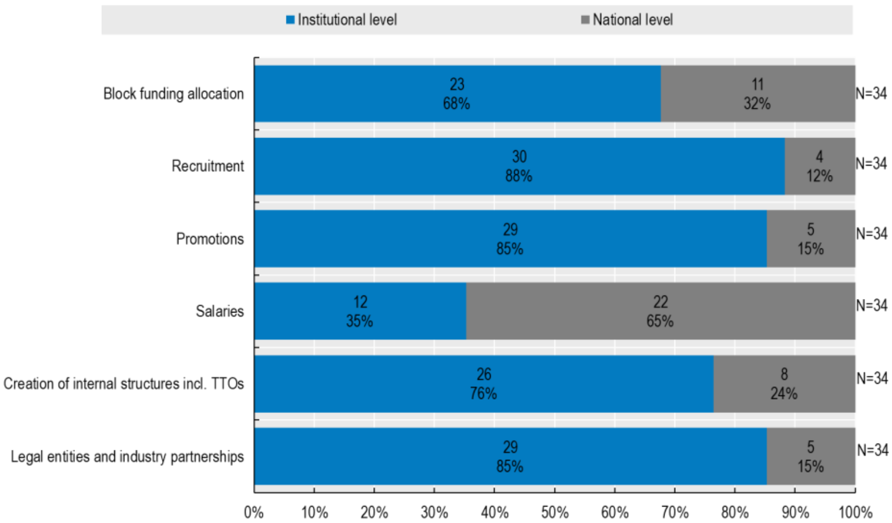
The extent to which HEIS and PRIs are autonomous in taking decisions over their budget, human resources and industry relations shapes the role of these institutions in the design of research policy. Figure 11 shows that HEIs (Panel A) and PRIs (Panel B) are very autonomous across the OECD with PRIs having autonomy in more countries across all dimensions.

Regarding industry relations, Panel A shows that 26 of 34 countries (or 76%) report HEIs can decide about the creation of academic departments (e.g. research centres in specific fields) and functional units (e.g. technology transfer offices), and in 29 of 34 countries (85%) they can create legal entities (e.g. for-profit spin-offs) and industry partnerships (e.g. joint R&D units). A similar picture emerges for PRIs (Panel B). 24 of 27 countries (or 89%) report PRIs can decide about the creation of academic departments and functional units such as technology transfer offices, and in 25 of 27 countries (93%) PRIs can create legal entities and industry partnerships.

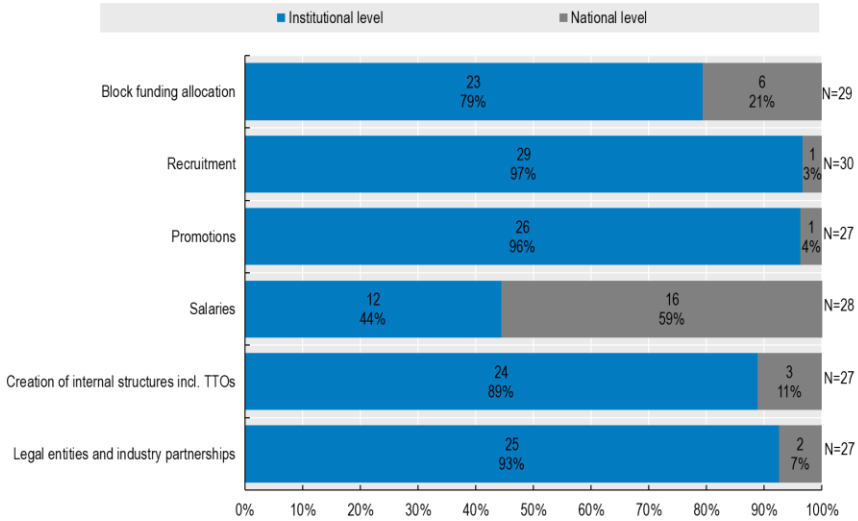
In terms of budget decisions, Panel A shows that in 23 of 34 (68%) OECD countries, public HEIs decide themselves about allocations of institutional block funding to teaching, research, and innovation activities. In the other 11 countries (32%), national laws or guidelines restrict the internal allocation of funds across budget items, such as e.g. personnel costs and capital expenditures. Regarding PRIs, in 23 of 29 (79%) countries PRIs can allocate their budget allocations freely while 6 countries (21%) report nationally binding laws or guidelines (Panel B).

Figure 11. HEIs' and PRIs' autonomy across the OECD-35

Panel A: Autonomy of HEIs



Panel B: Autonomy of PRIs



Note: The figure corresponds to question 3.4.a and b ("Who decides about allocations of institutional block funding for teaching, research and innovation activities in HEIs\PRIs?"), 3.5.a and b ("Who decides about recruitment of academic staff in HEIs\PRIs?"), 3.5.c and d ("Who decides about salaries of academic staff in HEIs\PRIs?"), 3.5.e and f ("Who decides about reassignments and promotions of academic staff in HEIs\PRIs?"), 3.6.a and b ("Who decides about the creation of academic departments, such as research centres in specific fields, and functional units, e.g. technology transfer offices, in HEIs\PRIs?"), and 3.6.c and d ("Who decides about the creation of legal entities and industry partnerships in HEIs? \PRIs"). Information on HEIs' autonomy is missing for New Zealand. Information on PRIs' autonomy is missing for Australia, Iceland, Japan, New Zealand, and the United States. Information on PRIs' autonomy with regard to block funding allocations is missing for the United Kingdom. Information on PRIs' autonomy with regard to recruitment is missing for Turkey. Information on PRIs' autonomy with regard to salaries and promotions is missing for Estonia, Luxembourg and Turkey. Information on PRIs' autonomy with regard to internal structures and legal entities is missing for the Netherlands, Turkey, and the United Kingdom.
Source: Data on university autonomy for Austria, Czech Republic, Denmark, Estonia, Finland, France, Germany (Brandenburg, Hesse and North Rhine Westphalia), Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, the Netherlands, Norway, Poland, Portugal, Slovak Republic, Spain, Sweden, Switzerland, Turkey and the United Kingdom is based on a survey conducted by the European University Association between 2010 and 2011. The answers were provided by Secretaries General of national rectors' conferences and can be found in the report by the European University Association (Estermann et al., 2015) at www.eua.be/Libraries/publications/University_Autonomy_in_Europe_II_-_The_Scorecard.pdf?sfvrsn=2.
Data on university autonomy for Australia, Canada, Chile, Israel, Japan, Korea, Mexico, New Zealand, and the United States as well as data on PRIs' autonomy was collected by authors using a survey.

Looking specifically at federal systems, HEIs are free to allocate institutional block funding to different activities. In Canada, provinces and territories are responsible to provide base funding to HEIs, and how this is allocated internally for teaching and other purposes varies from province to province, territory to territory and institution to institution. The picture is more complex for PRIs in federal systems. In Canada, Germany, Switzerland and the United States, leading PRIs are mostly funded by the Federal governments while Belgium and Spain have national and regional PRIs that receive respectively funding from the national governments and regional government.

HEIs have gained more independence from national ministries in a number of countries. In France, HEIs are free to establish their own for-profit entities and joint R&D with industry since 2011 (Freedom and Responsibilities for Universities Act 2011). In Portugal, HEIs were granted more autonomy by the Law 62/2007 of 10 September 2007 on Higher

Education Institutes (RJIES). In Austria, they gained full autonomy over financial and organisational affairs in 2002 (University Act).

In terms of hiring and promoting staff, public HEIs are free to decide about recruitment of academic staff in 30 of 34 (88%) OECD countries, and in 29 of 34 countries (or 85%) they decide freely about promotions (Panel A). PRIs are free to decide about recruitment in 29 of 30 countries (97%), and about promotions in 26 of 27 countries (96%) (Panel B).

HEIs and PRIs enjoy less autonomy with regard to salaries: Panel A shows that only in 12 of 35 countries (or 35%) HEIs can freely set the salaries of academic staff while Panel B shows that PRIs are free to decide about the level of salaries in 12 of 34 countries (44%). The way these salaries are set differs quite substantially. In some countries national laws set salary bands for academic staff, while others have collective bargaining arrangements in place at the national level (Box 4).

Box 4. Examples of decentralisation of decisions about salaries of academic staff

Sweden: HEIs and PRIs in Sweden are free to hire academic staff, to set salaries, and they can promote and reassign staff freely. National guidelines and laws exist that require institutions to publish open positions or the composition of the selection committee. In Sweden, professorial vacancies generally have to be publicly announced. However, a vacancy may be filled without public notification if a person holding very specific and unique competencies is directly invited to take up the post or if a post is filled for a fixed period only.

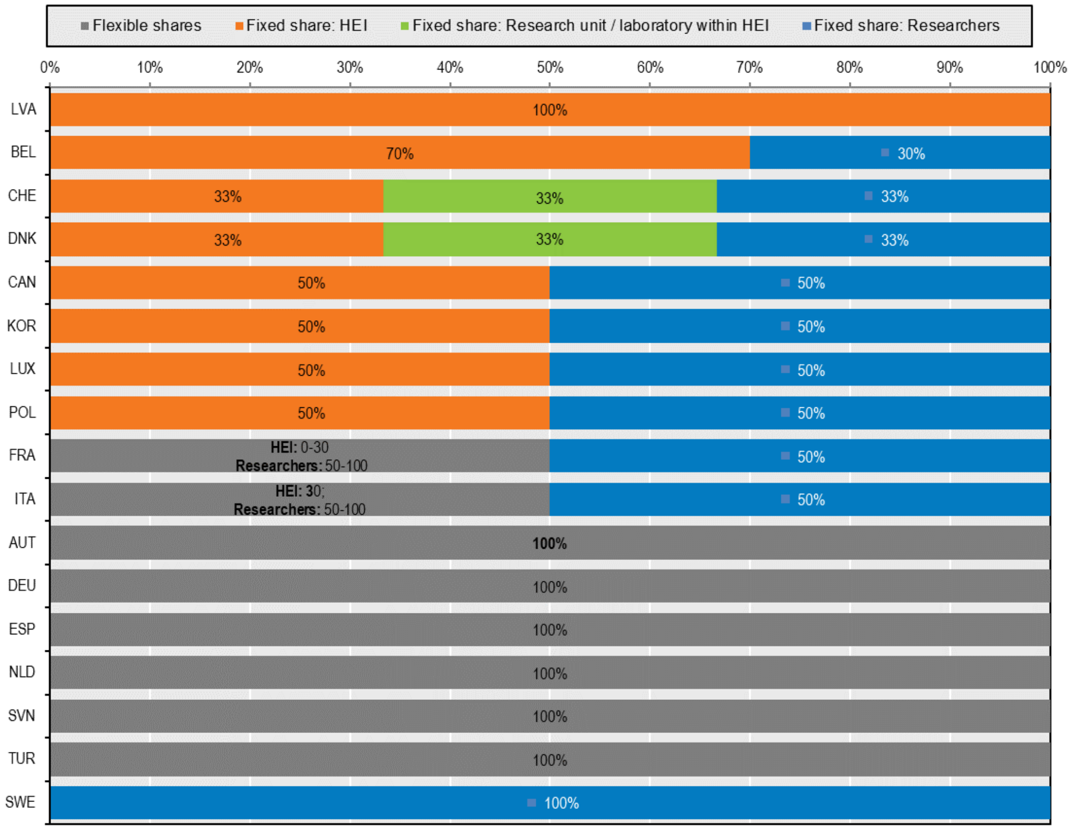
UK: With regard to the decisions about salaries of academic staff at HEIs and PRIs, national guidelines exist and salaries for all senior academic staff are negotiated with other parties. For most academic staff, universities tend to adhere to a nationally agreed 51-point pay scale. Imperial College and the University of Northumbria, which do not follow the scale, have nevertheless adopted similar pay structures. It is important to note that these are only guidelines. Within the minimum and maximum salaries, universities have autonomy over their salary grading structure. For example, an institution may choose to use ten or 20 instead of 51 points, and, crucially, it can freely decide on the appropriate qualifying criteria for the various scales. In addition, the nationally agreed pay scale normally does not apply to the professorial level or to senior management staff, including the executive head. In these cases, salaries above the GBP 58 258 maximum mark can be determined freely by the institution.

Source: Estermann et al. (2015).

Countries also differ with regard to revenues that researchers may receive from intellectual property (IP) at HEIs, such as patent license income. In 16 of 33 countries for which information is available, HEIs themselves set those revenue schemes and there exist a considerable degree of variation across universities. In 17 countries, national guidelines set the share that researchers receive (Figure 12). In Denmark, common practice across institutions establishes revenue sharing of 30% for researchers, 30% for the research unit or laboratory within the HEI/PRI, and 30% for the HEIs/PRIs. Where national guidelines exist, the share for researchers is often set between 33% and 50%. Sweden is a notable exception and grants 100%, a practice referred to as professor’s privilege. In contrast, 100% of revenue accrues to universities in Latvia.

Some changes to revenue schemes are fairly recent. Turkey just introduced the new Industrial Property Law (Law No. 6769 of 10 January 2017) that designates ownership of IP from publicly funded research to universities. Researchers are entitled to receive at least one third of the revenue from revenues of their IP. Before the reforms the revenue shares were not regulated.

Figure 12. Who earns what share of revenues stemming from IP created from publicly funded research at HEIs?



Note: This figure corresponds to question 3.7.a ("Who earns what share of revenues stemming from IP (patents, trademarks, design rights, etc.) created from publicly funded research in HEIs?"). The countries reported have national obligations or guidelines in place on revenue shares.

HEIs set their own schemes in Australia, Chile, the Czech Republic, Estonia, Finland, Greece, Hungary, Ireland, Israel, Japan, Mexico, Norway, Portugal, the Slovak Republic, the United Kingdom, and the United States. Information was not available for Iceland and New Zealand.

4. CONCLUSION

This paper provides a first systematic mapping of the governance of public research across all 35 OECD countries for the period 2005-17. The mapping describes the institutional arrangements that define how various governmental bodies (ministries, councils, and agencies), HEIs, PRIs and private actors are involved in the design and implementation of policies regarding publicly-funded research conducted in HEIs and PRIs. A number of interesting perspectives on differences in the governance and on trends across the OECD emerge, raising questions for further more in-depth analysis. In particular, exploring the database to investigate the determinants of governance structures and their impacts, for instance by using cross-country regression frameworks, would be interesting. Another application of the data is to investigate how governance relates to the performance of HEIs and PRIs. Moreover, developing a more comprehensive set of indicators of research and innovation policies will also be valuable.

ENDNOTES

1 The information provided here and in a few other sections is for fewer than 35 OECD countries because of missing information.

2 Information on block grants for public research institutes in Iceland is missing.

3 In Switzerland, the federal government is in charge of the supervision of the federal HEIs (Eidgenössische Technische Hochschulen, ETHs), while the cantons are in charge of the supervision of the cantonal universities. Both have equal rights in their respective domains. Cantons have their own governance structures, which are in general comparable to the national structures.

4 In Germany, while the Innovation Dialogue specialises on information and communication technologies (ICT) and digitalisation, the Expert Commission for Research and Innovation is responsible for innovation policy advice. The Council of Science and Humanities specialises on higher education and research policy. The characterisation of the functions of councils below focuses on the main council in charge (i.e. in the case of Germany the Council of Science and Humanities is characterised).

5 Israel does not have a national STI strategy in place but uses a series of national reports and STI related policy documents in specific scientific research, technologies and economic fields to set priorities. These address societal challenges (demographic change, digital transformation, the green economy, and health) and specific technologies (energy technologies, ICT, and nanotechnologies). In Luxembourg, several sector-specific priority lists and strategies are in place.

REFERENCES

Aghion, P., Dewatripont, M., Hoxby, C., Mas-Colell, A., Sapir, A., and B. Jacobs (2010), "The governance and performance of universities: Evidence from Europe and the US", *Economic Policy*, Vol. 25, pp. 7-59.

Alemani, E., Klein, C., Koske, I., Vitale, C., and I. Wanner, (2013), "New indicators of competition law and policy in 2013 for OECD and non-OECD Countries", *OECD Economics Department Working Papers*, No. 1104, OECD Publishing, Paris, https://doi.org/10.1787/5k3ttg4r657h-en.

Austrian Science Council (2016), "Analyse der Leistungsvereinbarungen 2016–2018 und Empfehlungen", (Analysis of the Performance Agreements 2016-2018 and Recommendations), Vienna, Austria, http://www.wissenschaftsrat.ac.at/news/Endversion_Leistungsvereinbarungen%202016_2018.pdf (accessed 16 February 2017).

Borrás, S., and J. Edler (2014), "Introduction: On Governance, Systems and Change", in Borrás, S., and Edler, J. (eds.), *The Governance of Socio-Technical Systems: Explaining Change*, pp. 1-22, Edward Elgar Publishing, Cheltenham.

Borrás, S., and C. Edquist (2013), "The choice of innovation policy instruments", *Technological Forecasting and Social Change*, Vol. 80, pp. 1513-1522.

Breznitz, D. (2007), *Innovation and the State*, Yale University Press, New Haven.

Center for Higher Education Policy Studies (2015), "Performance-based funding and performance agreements in fourteen higher education systems", pp. 111-122 and pp. 127-148, Enschede, http://doc.utwente.nl/93619/7/jongbloed%20ea%20performance-based-funding-and-performance-agreements-in-fourteen-higher-education-systems.pdf (accessed 18 October 2016).

De Boer, H., B. Jongbloed, P. Benneworth, L. Cremonini, R. Kolster, A. Kottmann, K. Lemmens-Krug, and H. Vossensteyn (2015), "Performance-based Funding and Performance Agreements in Fourteen Higher Education Systems: Report for the Ministry of Education, Culture and Science", Center for Higher Education Policy Studies, No. C15HdB014I, Enschede, http://doc.utwente.nl/93619/7/jongbloed%20ea%20performance-based-funding-and-performance-agreements-in-fourteen-higher-education-systems.pdf (accessed 05 October 2016).

Delegation of the European Union to Israel (2016), "Scientific cooperation", website, http://eeas.europa.eu/archives/delegations/israel/eu_israel/scientific_cooperation/index_en.htm (accessed 16 November 2016).

Department for Business, Energy and Industrial Strategy Consultation Hub (2017), website, https://beisgovuk.citizenspace.com/ (accessed 07 May 2017).

Department for Education Consultation Hub (2016), website, https://consult.education.gov.uk/ (accessed 22 October 2016).

Dutch National Research Agenda (2016), "From Vision for Science to Dutch National Research Agenda in 365 days", website, www.wetenschapsagenda.nl/publicatie/from-vision-for-science-to-dutch-national-research-agenda-in-365-days/?lang=en (accessed 18 November 2016).

Estermann, T., T. Nokkala, M., and M., Steinel (2015), "University Autonomy in Europe II The Scorecard. Brussels: European University Association", http://www.eua.be/Libraries/publications/University_Autonomy_in_Europe_II_-_The_Scorecard.pdf?sfvrsn=2 (accessed 19 September 2016).

European Commission (2016a), "European Structural and Investment Funds", website, http://ec.europa.eu/contracts_grants/funds_en.htm (accessed 16 November 2016).

European Commission (2016b), "Horizon 2020 – the EU Framework Programme for Research and Innovation", website, https://ec.europa.eu/programmes/horizon2020/en/ (accessed 16 November 2016).

European Commission (2016c), "Registered countries and regions in the S3 Platform", website, http://s3platform.jrc.ec.europa.eu/s3-platform-registered-regions (accessed 16 November 2016).

European Commission (2016d), "Rio Country Report 2015: Slovak Republic", p 14, Report EUR 27860 EN, Joint Research Centre, Institute for Prospective Technological Studies, Sevilla, Spain, https://rio.jrc.ec.europa.eu/en/country-analysis/Slovakia/country-report (accessed 08 March 2017).

European Commission (2016e), "Smart Specialisation", website, https://ec.europa.eu/jrc/en/research-topic/smart-specialisation (accessed 16 November 2016).

European Commission (2015a), "EU 2020 Targets", website, http://ec.europa.eu/europe2020/europe-2020-in-a-nutshell/targets/index_en.htm (accessed 16 November 2016).

European Commission (2015b), "RIO Country Report Belgium 2014", Report EUR 27294 EN, Joint Research Centre, Institute for Prospective Technological Studies, Sevilla, Spain, https://rio.jrc.ec.europa.eu/sites/default/files/riowatch_country_report/RIO%20Country%20Report%202014_Belgium.pdf (accessed 02 February 2017).

European Commission (2014), Turkey joins Horizon 2020 research and innovation programme, website, http://europa.eu/rapid/press-release_IP-14-631_en.htm (accessed 05 May 2017).

Fuchs, E.R.H. (2010), "Rethinking the role of the state in technology development: DARPA and the case for embedded network governance", *Research Policy*, Vol. 39, pp. 1133-1147.

Ginarte, J.C., and W.G., Park, (1997), "Determinants of patent rights: A cross-national study", *Research Policy*, Vol. (26), pp. 283-301.

Grosso, G.M., Gonzales, F., Miroudot, S., Nordås, H.K., Rouzet, D., and A. Uenoi (2015), "Services in trade restrictiveness index (STRI): Scoring and weighting methodology", *OECD Trade Policy Papers*, No. 177, OECD Publishing, Paris, http://dx.doi.org/10.1787/5js7n8wbtk9r-en.

Guellec and Paunov (2018), "Innovation Policies in the Digital Age", in *OECD Science, Technology, and Innovation Outlook 2018*, OECD Publishing, Paris, forthcoming.

Koske, I., Wanner, I., Bitetti, R., and O. Barbiero (2015), "The 2013 update of the OECD product market regulation indicators: policy insights for OECD and non-OECD countries", *OECD Economics Department Working Papers*, No. 1200. DOI: http://dx.doi.org/10.1787/18151973.

OECD (2018), "STIP Compass", *European Commission OECD STI Compass: International Database on STI Policies* (database), https://stip.oecd.org/stip.html (accessed 19 July 2018).

OECD (2016a), *OECD Science, Technology and Innovation Outlook 2016*, OECD Publishing, Paris. http://dx.doi.org/10.1787/25186167.

OECD (2016b), *OECD Reviews of Innovation Policy: Sweden 2016*, OECD Publishing, Paris, http://dx.doi.org/10.1787/9789264250000-en.

OECD (2016c), *OECD Reviews of Innovation Policy: Luxembourg 2016*, OECD Publishing, Paris, http://dx.doi.org/10.1787/9789264232297-en.

OECD (2015), *Frascati Manual: Guidelines for Collecting and Reporting Data on Research and Experimental Development*, Paris, OECD Publishing, http://dx.doi.org/10.1787/9789264239012-en.

OECD (2014a), *OECD Reviews of Innovation Policy: France 2014*, OECD Publishing, Paris. http://dx.doi.org/10.1787/9789264214026-en.

OECD (2014b), *OECD Reviews of Innovation Policy: Netherlands 2014*, OECD Publishing, Paris. http://dx.doi.org/10.1787/9789264213159-en.

OECD (2011), *Public Research Institutions: Mapping Sector Trends*, OECD Publishing, Paris. http://dx.doi.org/10.1787/9789264119505-en.

Park, W.G. (2008), “International patent protection: 1960-2005”, *Research Policy*, Vol. 37, pp. 761-766.

Scottish Government Consultation Hub (2017), website, https://consult.scotland.gov.uk/ (accessed 7 May 2017).

SoSP (2016), About the Interagency Working Group on Science of Science Policy, website, http://www.scienceofsciencepolicy.net/page/about-interagency-working-group-science-science-policy-sosp-iwg (accessed 27 October 2016).

Stampfer (2017), “Performance Agreements in Austria: A Case Of Simultaneous Under- And Over-Steering”, in Austrian Council for Research and Technology Development (ed.), *Prospects and Future Tasks of Universities*, pp. 161-173, Lit Verlag, Zürich.

The Norwegian Mission to the EU (2016), Norway’s participation in EU programmes and agencies, website www.eu-norway.org/eu/Coopperation-in-programmes-and-agencies/#.WCxKrMlUsXh (accessed 16 November 2016).

University of Vienna (2015), Performance agreement of the University of Vienna (Leistungsvereinbarung 2016–2018), website (German), https://www.univie.ac.at/mtbl02/02_pdf/20151222.pdf (accessed 22 November 2016).

World Bank (2018), *Doing Business 2018: Reforming to Create Jobs*, World Bank Group, Washington.

ANNEX: METHODOLOGY

A.1. Ontology development and policy mapping across OECD countries

A.1.1. Ontology development

The mapping of the governance of public research presented in this paper is based on the development of an ontology of the governance of public research. The set of indicator questions included in the ontology were evaluated on the basis of complying with the following 4 criteria, which was adapted from the comparable policy analysis of Alemani, E., et al. (2013):

1. Analytical soundness/coverage: Clearly defined and understandable questions that characterise core dimensions of the governance of public research
2. Coverage: Questions allowing for cross-country responses, excluding questions on rarely used policy approaches adopted by few countries only
3. Measurability: Objective and fact-based questions for clearly specified time periods, excluding questions on respondents' perceptions
4. Explanatory value: Questions capture key features of country policy approaches at sufficient level of granularity to describe different policy practice where they exist

The ontology was developed based on an in-depth review of the relevant literature on the governance of public research, resulting in the selection of 3 policy dimensions: i) institutions in charge of priority setting, budget allocation and evaluation (e.g. Breznitz, 2007; Fuchs, 2010; Borrás and Edquist 2013), ii) policy co-ordination mechanisms (e.g. Borras and Edler, 2014) and iii) external stakeholder involvement and university autonomy (e.g. Aghion et al., 2010).

An intensive consultation and testing process was used to obtain the final ontology to ensure all indicator questions complied with the 4 criteria. A draft ontology was validated by country delegates and experts of the OECD Technology and Innovation Policy (TIP) Working Party, resulting also in the development of a common taxonomy for analytical soundness/coverage (see A.3).

On the basis of comments received, a revised questionnaire was used to gather information for a selection of countries. The exercise allowed identifying questions that were in some country contexts difficult to respond to, leading to a risk of subjective responses. Some questions did not allow capturing cross-country differences.

A.1.2. Policy Mapping

Data on country policies was collected by means of an indicator questionnaire (presented in section A.3 of this Annex). Data sources used for collecting the policy information include the European Commission/OECD International Survey on Science, Technology and Innovation Policies (and its predecessor, the OECD Policy Questionnaire on the STI Outlook (OECD, 2016a) and the OECD Country Innovation Policy Reviews (OECD, 2014a; 2014b; 2016b; 2016c). Where additional information was needed, national legislative texts (including labour laws, decrees and other legal documents), strategic policy documents (such as strategic frameworks), statements of purpose and other such official documents (such as statements on the status of councils, committees, etc.) were used.

The policy information was validated in extensive consultation with national delegates of the OECD Committee of Science and Technology Policy (CSTP), delegates of the OECD Technology and Innovation Policy (TIP) Working Party, and national authorities between March 2017 and March 2018. Delegations from 29 OECD countries provided detailed feedback: Australia, Austria, Belgium, Canada, Chile, the Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Israel, Italy, Japan, Korea, Latvia, Mexico, the Netherlands, Norway, Poland, Portugal, the Slovak Republic, Slovenia, Spain, Sweden, Switzerland, and the United Kingdom. Detailed feedback for Luxembourg was provided by a country expert who participated in the OECD Innovation policy Review of Luxembourg in 2016.

A.2. Questionnaire

A.2.1. Dimension 1: Institutions in charge of priority setting, budget allocation and evaluation

Dimension 1 covers the institutions in charge of priority setting, budget allocation and evaluation (vertical governance). Governance can be "top-down" where national ministries and agencies target government-based research and firm subsidies, or take the form of "bottom-up" decentralized arrangements that where agencies, councils, and associations of universities and public research institutions decide among themselves (Breznitz, 2007; Fuchs, 2010). Funding arrangements also form part of vertical governance, i.e. performance contracts that bind institutional funding to pre-defined goals with the ministry/agency in charge.

Table A1. Questions on institutions in charge of priority setting, budget allocation and evaluation

Institutions in charge of priority setting								
Q.1.1. Who mainly decides on the scientific, sectoral and/or thematic priorities of HEIs and PRIs? * This is also the case when there is no top-down priority setting by ministries or funding agencies and HEIs, PRIs, and research groups apply for research and innovation funding through open calls (e.g. Switzerland, Sweden).	**National ministry** (e.g. Ministry of Education, Higher Education and Research)	**National agency** (e.g. Funding Agency for Innovation)	**Research and Innovation Council/Committee**	**Federal state and/or regional ministry** (e.g. Bavarian State Ministry of Education and Cultural Affairs, Research and Culture)	**Federal state and/or regional agencies**(e.g. Research Foundation Flanders FWO)	**Institutions themselves*** (i.e. institutions decide on programme priorities and the State transfers funds directly to institutions.)	Not applicable	**Source:**
	Check the relevant boxes [multiple responses not possible, check only the main body]							

a) Decisions on scientific, sectoral and/or thematic priorities of budget allocations for HEIs Please provide further information if responses differ for research and innovation programmes.								
a_1) *Name of institution(s)*								
b) Decisions on scientific, sectoral and/or thematic priorities of budget allocations for PRIs								
b_1) *Name of institution(s)*								
c) Which are the main mechanisms in place to decide on scientific, sectoral and/or thematic priorities of national importance, e.g. digital transition, sustainability? Please describe who is involved and who decides on the priorities (e.g., government, research and innovation councils, sector-specific platforms including industry and science, etc.). *Scientific priorities* refer to scientific disciplines, e.g. biotechnology; *Sectoral priorities* refer to industries, e.g. pharmaceuticals; *Thematic priorities* refer to broader social themes, e.g. digital transition, sustainability, etc.								
d) *From 2005-17, were any significant changes introduced as to how decisions on scientific, sectoral and/or thematic orientation of major programmes are taken? (e.g. establishment of agencies that decide on content of programmes)*	*Please indicate these reform(s)*						*No changes made*	**Source:**

	Institutions in charge of budget allocation							
Q.1.2. *Who allocates public funding to HEIs and PRIs?* *Allocations* are direct transfers to HEIs and PRIs once the public budget has been approved and transferred to the responsible ministries and/or funding agencies.	**National ministry** (e.g. Ministry of Education, Higher Education and Research)	**National agency** (e.g. Funding Agency for Innovation)	**Research and Innovation Council/Committee**	**Federal state and/or regional ministries** (e.g. Bavarian State Ministry of Education and Cultural Affairs, Research and Culture)	**Federal state and/or regional agencies** (e.g. Research Foundation Flanders FWO)	**Institutions themselves** (i.e. State transfers budget directly to institutions and they are responsible for programme development and budget allocations)	Not applicable	**Source:**
	Check the relevant boxes [multiple responses not possible, check only the main body]							
a) *Institutional block funding for HEIs* *Institutional block* funds (or to general university funds) support institutions and are usually transferred directly from the government budget.								
a_1) *Name of institution(s)*								
b) *Institutional block funding for PRIs*								
b_1) *Name of institution(s*								

<table>
<tr><td>c) Project-based funding of research and/or innovation for HEIs and PRIs

Project-based funding provides support for research and innovation activities on the basis of competitive bids.
Funding for research supports research personnel, basic and applied research activities, research infrastructure and/or equipment.
Funding for innovation supports technology development, research commercialisation, technology transfer, research commercialisation, and/or joint R&D between PRIs and industry.</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>c_1) Name of institution(s)</td><td colspan="8"></td></tr>
<tr><td rowspan="2">d) Is there a transnational body that provides funding to HEIs and PRIs (e.g. the European Research Council)?</td><td colspan="3">Yes</td><td colspan="4">No</td><td>Source:</td></tr>
<tr><td colspan="3"></td><td colspan="4"></td><td></td></tr>
<tr><td>d_1) Name of transnational body that provides funding to HEIs and PRIs</td><td colspan="8"></td></tr>
<tr><td>e) What is the importance of such funding relative to national funding support? (Share of public R&D funding)</td><td colspan="8"></td></tr>
<tr><td rowspan="2">f) From 2005-17, were any changes made to way programmes are developed and funding is allocated to HEIs and PRIs?
(e.g. merger of agencies, devolution of programme management from ministries to agencies)</td><td colspan="6">Please indicate which reform(s)</td><td>No changes made</td><td>Source:</td></tr>
<tr><td colspan="6"></td><td></td><td></td></tr>
</table>

Q.1.3. Do performance contracts determine funding of HEIs*? * In case performance contracts are in place that bind funding of PRIs, please provide information about them. *Institutional block* funds can be partly or wholly distributed based on performance. *Performance contracts* define goals agreed between ministry/agency and HEIs/PRIs and link it to *future* block funding of HEIs and PRIs.	*Yes*			*No*	**Source:**
a) *Institutional block funding of HEIs?*					
b) *Share of HEI budget subject to performance contract*	*Please indicate percentage*			*No performance contracts exist*	**Source:**
c) *Do performance contracts include quantitative indicators for monitoring and evaluation?*	*Yes*		*No*	*No performance contracts exist*	**Source:**
d) *Which, if any, performance aside from research and education is set out in performance contracts?*	*Innovation-related*	*Socio-economic challenges*	*Local economy support*	*No performance contracts exist*	**Source:**
d_1) *Please specify which performance indicators…*					
e) *Do HEIs participate in the formulation of main priorities and criteria used in performance contracts?*	*Yes*		*No*	*No performance contracts exist*	**Source:**
f) *Do the same priorities and criteria set in performance contracts apply to all HEIs?*	*All HEIs*		*Individual HEIs*	*No performance contracts exist*	**Source:**
g) *Are any other mechanisms in place to allocate funding to HEIs and PRIs? Please specify…*					

***h)** From 2005-17, were any changes made to funding of HEIs and PRIs?*	*Please indicate which reform(s)*	*No changes made*	**Source:**

Institutions in charge of evaluation

Q.1.4. Who decides on the following key evaluation criteria of HEIs and PRIs?	**National ministry** (e.g. Ministry of Education, Higher Education and Research)	**National agency** (e.g. Funding Agency for Innovation)	**Research and Innovation Council/Committee**	**Federal state and/or regional ministries**	**Federal states and/or regional agencies** (e.g. Research Foundation Flanders FWO)	**Institutions themselves**	**Others** (e.g. external consultancy firms)	Not applicable	**Source:**
	Check (X) the relevant boxes [multiple responses not possible, check only the main body]								
***a)** Setting criteria to use when evaluating performance of HEIs*									
***a_1)** Name of institution(s)*									
***b)** Evaluating HEIs' performance*									
***b_1)** Name of institution(s)*									
***c)** Monitoring HEIs' performance*									
***c_1)** Name of institution(s)*									
***d)** Setting criteria to use when evaluating performance of PRIs*									
***d_1)** Name of institution(s)*									
***e)** Evaluating PRIs' performance*									
***e_1)** Name of institution(s)*									
***f)** Monitoring PRIs' performance*									
***f_1)** Name of institution(s)*									

***g)** From 2005-17, was any institution created for evaluating HEIs and PRIs or were any changes made to criteria applied for evaluations of HEIs and PRIs?*	*Please indicate which institutions or reform(s)*	*No changes made*	**Source for answer:**

Q.1.5. *Which recent reforms to vertical governance structures of HEIs and PRIs were particularly important?*	

Comments:
a) In case of multiple levels being responsible for the above listed policy-making decisions, indicate which institutions have higher responsibilities than others, e.g., final decision making power or veto power?
b) Please indicate if any of the answers above are difficult to provide and, if so, why (e.g. because governance is shared by several institutions).
c) Please indicate if different Federal States/regions within a country have their own vertical governance arrangements and how these differ.

A.2.2. Dimension 2: Co-ordination of policy

Dimension 2 describes policy co-ordination (or "whole-of-government" approaches i.e. a variety of institutional arrangements that ensure that different actors involved in research policy are co-ordinated. Research and innovation councils/committees may mediate between different ministries and agencies, provide policy advice, set policy priorities and/or oversee policy evaluation. National research and innovation strategies or plans may also serve policy co-ordination. At the operational level, programmes that are jointly managed by various agencies may foster coordination between agencies. Agencies themselves may also take on a coordinating role in the system.

Table A2. Questions on co-ordination of policy

Research and Innovation Councils			
Q.2.1. Research and Innovation Councils/Committees	*Yes*	*No*	**Source:**
a) *Is there a Research and Innovation Council, i.e. non-temporary public body that takes decisions concerning HEI and PRI policy, and that has explicit mandates by law or in its statutes to either* ▪ *provide policy advice (i.e. produce reports);* ▪ *and/or oversee policy evaluation;* ▪ *and/or coordinate policy areas relevant to public research (e.g. across ministries and agencies);* ▪ *and/or set policy priorities (i.e. strategy development, policy guidelines);* ▪ *and/or joint policy planning (e.g. joint cross-ministry preparation of budgetary allocations)?*			
b) *What is the name of the main research and/or innovation Council/Committee?*	*Name of institution(s)*		*No such body/bodies exist(s)*

c) *Are there any other research Councils/Committees?*				
Q.2.2. With reference to Q.2.1.b., does the Council's mandate explicitly include…	*Yes*	*No*	*No such body/bodies exist(s)*	**Source:**
	Please check (X) the relevant boxes			
a) *Policy coordination* (e.g. coordinating inter-governmental activities and policy domains)				
b) *Preparation of strategic priorities*				
c) *Decision-making on budgetary allocations*				
d) *Evaluation of policies' implementation (including their enforcement)*				
e) *Provision of policy advice*				
Q.2.3. With reference to Q.2.1.b., who formally participates in the Council?	*Yes*	*No*	*No such body/bodies exist(s)*	**Source:**
	Please check (X) the relevant boxes			
a) *Head of State*				
b) *Ministers*				
c) *Government officials* (civil servants and other representatives of ministries, agencies and implementing bodies)				
d) *Funding agency representatives*				
e) *Local and regional government representatives*				
f) *HEI representatives*				
g) *PRI representatives*				
h) *Private sector*				
i) *Civil society*				
j) *Foreign experts*				
Q.2.4. With reference to Q.2.1.b., does the Council have …	*Yes*	*No*	*No such body/bodies exist(s)*	**Source:**
a) *Its own staff?*				
a_1) *If so, please indicate number of staff…*				
b) *Its own budget?*				
b_1) *If so, please indicate amount of annual budget available…*				
c) *From 2005-17, were any changes made to the mandate of the Council, its functions, the composition of the Council, the budget and/or the Council's secretariat? Was the Council created during the time period?*	*Please indicate which reform(s)*	*No changes to Council made*	**Source:**	

STI Strategy or Plan

Q.2.5. National STI strategies.	*Yes*	*No*	**Source:**
a) Is there a national non-sectoral STI strategy or plan?			
b) What is the name of the main national STI strategy or plan?	*Name of STI strategy*	*No such strategy exists*	

Q.2.6. Does the national STI strategy or plan address any of the following priorities? Specify if another more dedicated strategy (e.g. a specific plan) covers these topics? *Please refer to the main STI strategy. If additional strategies address the following issues, please provide further information on them.*	*Please check (X) the relevant boxes*	*There is no such strategic framework*	**Source:**
***a)** Specific themes and/or societal challenges* (e.g. Industry 4.0; "green innovation"; health; environment; demographic change and wellbeing; efficient energy; climate action)			

***a_1)** Which of the following themes and/or societal challenges are addressed?*	**Demographic change** (i.e. ageing populations, etc.)	**Digital economy** (e.g. big data, digitalisation, industry 4.0)	**Green economy** (e.g. natural resources, energy, environment, climate change)	**Health** (e.g. Bioeconomy, life science)	**Mobility** (e.g. transport, smart integrated transport systems, e-mobility)	**Smart cities** (e.g. sustainable urban systems urban development)	*No such strategy exists*	**Source:**
	Please check (X) the relevant boxes							
***a_2)** Please specify which priorities …*								

***b)** Specific scientific research, technologies and economic fields* (e.g. ICT; nanotechnologies; biotechnology)			

***b_1)** Which of the following scientific research, technologies and economic fields are addressed?*	**Agriculture and agricultural technologies**	**Energy and energy technologies** (e.g. energy storage, environmental technologies)	**Health and life sciences** (e.g. biotechnology, medical technologies)	**ICT** (e.g. big data, digital platforms, data privacy)	**Nanotechnology and advanced manufacturing** (e.g. robotics, autonomous systems)	*No such strategy exists*	**Source:**
	Please check (X) the relevant boxes						

***b_2)** Please specify which priorities …*				
***c)** Specific Federal States/regions* (e.g. smart specialisation strategies)				
***c_1)** Please specify which priorities…*				
***d)** Supranational or transnational objectives set by transnational institutions* (for instance related to European Horizon 2020)				
***d_1)** Please specify which priorities …*				
***e)** Quantitative targets for monitoring and evaluation* (e.g. setting as targets a certain level of R&D spending for public research etc.)				
***f)** From 2005-17, was any STI strategy introduced or were any changes made existing STI strategies?*	*Please indicate which reform(s)*	*No changes to strategic frameworks*	**Source:**	
Q.2.7. *What reforms to policy co-ordination regarding STI strategies and plans have had particular impact on public research policy?*				

Inter-agency programming and role of agencies and ministries

Q.2.8. Does inter-agency joint programming contribute to the co-ordination of HEI and PRI policy? *<u>Inter-agency joint programming</u> refers to formal arrangements that result in joint action by implementing agencies, such as e.g. sectoral funding programmes or other joint policy instrument initiatives between funding agencies*	*Yes*	*No*	**Source:**
Q.2.9.a) Is co-ordination within the mandate of agencies? What tasks are performed that help co-ordination? Please specify.			
***b)** From 2005-17, were any changes made to the mandates of agencies tasked with regards to inter-agency programming? Were new agencies created with the task to coordinate programming during the time period?*	*Please indicate reform(s)*	*No changes made*	**Source:**
Q.2.10. What reforms of the institutional context have had impacts on public research policy?			

Comments:
a) In the event of differences in the way the above listed co-ordination activities target public/private HEIs and PRIs, please indicate these differences.
b) Please indicate if any of the above answers are difficult to provide and, if so, why.

A.2.3. Dimension 3: Stakeholders consultation and institutional autonomy

Dimension 3 refers to external stakeholders – HEIs/PRIs themselves, the private sector, NGOs and citizens – can be involved to varying degree in governance, namely through their participation as formal members in research and innovation councils as well as in councils or governing boards of HEIs. Public consultation exercises (e.g. online consultation platforms) are a means to involve diverse stakeholders, including civil society. Autonomy of HEIs and PRIs refers to their operating freedom with regard to internal allocation of funding, human resources policies (including recruitment of academic staff, salaries, reassignments and promotions), and industry relations (including the creation of legal entities and industry partnerships and revenue sharing of IP transactions).

Table A3. Questions on stakeholder consultation

Participation in councils and university boards						
Q.3.1. Do stakeholders participate as formal members in the following key consultation processes?	*Private sector*	*Civil society (citizens/ NGOs/ foundations)*	*HEIs / PRIs and/or their associations*	*Foreign experts*	*No formal consultation*	**Source:**
a) *Research and Innovation Councils:* (i.e. Formal membership as provided by statutes of Council)						
a_1) *Please specify the type of stakeholder e.g. for private sector the name and whether it is a large enterprise, small and medium enterprise, industry association or other…*						
b) *Council/governing boards of HEIs* (i.e. Formal membership of <u>external</u> stakeholders as provided by statutes of HEIs)						

***b_1)** Please specify the type of stakeholder e.g. for private sector the name and whether it is a large enterprise, small and medium enterprise, industry association or other…*				
Online public consultations and temporary stakeholder consultation processes				
Q.3.2.a) Are there online consultation platforms in place to request inputs regarding HEI and PRI policy?	*Yes*	*No*	*Not applicable*	**Source:**
***b)** Which aspects do these online platforms address?* *Please provide examples*, e.g. open data, open science.				
***c)** From 2005-17, were any reforms made to widen inclusion of stakeholders and/or to improve consultations, including online platforms?*	*Please indicate which reform(s)*	*No changes made*		**Source:**
Q.3.3. *Which reforms to consultation processes have proven particularly important?*				

Comments: Indicate if any of the answers above is difficult to provide and, if so, why this is the case.

Table A4. Questions on institutional autonomy

Budget autonomy						
	National level (e.g. Ministry of National Education, Higher Education and Research; Funding Agency for Innovation)	**Federal states and/or regional level** (e.g. Bavarian State Ministry of Education and Cultural Affairs, Research and Culture)	**Institutions themselves** (e.g. governing body/council of HEIs/PRIs)	**Others** (e.g. departments within HEIs)	Not applicable	**Source:**
Q.3.4.a) *Who decides about allocations of institutional block funding for teaching, research and innovation activities in HEIs?* *National/regional level*: If HEIs face national constraints on using block funds, i.e. funds cannot be moved between categories such as teaching, research, infrastructure, operational costs, etc. This option						

also applies if the ministry pre-allocates budgets for universities to cost items, and HEIs are unable to distribute their funds between these. *Institutions themselves:* If HEIs are entirely free to use their block grants.						
***b)** Who decides about allocations of institutional block funding for teaching, research and innovation activities in PRIs?*						
		Human resources and autonomy				
Q.3.5.a) *Who decides about recruitment of academic staff in HEIs?* *National/regional level*: If recruitment needs to be confirmed by an external national/regional authority; if the number of posts is regulated by an external authority; or if candidates require prior accreditation. This option also applies if there are national/regional laws or guidelines regarding the selection procedure or basic qualifications for senior academic staff. *Institutions themselves*: If HEIs are free to hire academic staff. This option also applies to cases where laws or guidelines require the institutions to publish open positions or the composition of the selection committees which are not a constraint on the hiring decision itself.						
***b)** Who decides about recruitment of academic staff in PRIs?*						
***c)** Who decides about salaries of academic staff in HEIs?* *National/regional level*: If salary bands are negotiated with other parties, if national civil servant or public sector status/law applies; or if external authority sets salary bands. *Institutions themselves*: If HEIs are free to set salaries, except minimum wage.						
***d)** Who decides about salaries of academic staff in PRIs?*						

e) *Who decides about reassignments and promotions of academic staff in HEIs?* *National/regional level*: If promotions are only possible in case of an open post at a higher level; if a promotion committee whose composition is regulated by law has to approve the promotion; if there are requirements on minimum years of service in academia; if automatic promotions apply after certain years in office, or if there are promotion quotas. *Institutions themselves*: If HEIs can promote and reassign staff freely.						
f) *Who decides about reassignments and promotions of academic staff in PRIs?*						
Industry relations and autonomy						
Q.3.6.a) *Who decides about the creation of academic departments (such as research centres in specific fields) and functional units (e.g. technology transfer offices) in HEIs?* *National/regional level*: If there are national guidelines or laws on the competencies, names, or governing bodies of internal structures, such as departments or if prior accreditation is required for the opening, closure, restructuring of departments, faculties, technology offices, etc. *Institutions themselves*: If HEIs are free to determine internal structures, including the opening, closure, restructuring of departments, faculties, technology offices, etc.						
b) *Who decides about the creation of academic departments (such as research centres in specific fields) and functional units (e.g. technology transfer offices) in PRIs?*						

c) *Who decides about the creation of legal entities and industry partnerships in HEIs?* *National/regional level*: If there are restrictions on legal entities, including opening, closure, and restructuring thereof; if restrictions apply on profit and scope of activity of non-profit organisations, for-profit spin-offs, joint R&D, etc. *Institutions themselves:* If HEIs are free to create non-profit organisations, for-profit spin-offs, joint R&D, etc.						
d) *Who decides about the creation of legal entities and industry partnerships in PRIs?*						

Q.3.7.a) Who earns what share of revenues stemming from IP (patents, trademarks, design rights, etc.) created from publicly funded research in HEIs?	*Please indicate the relative percentage (confirm that the percentages add up to 100%)*	*HEIs set their own schemes*	**Source:**
HEI			
Research unit / laboratory within HEI			
Researchers			
Other			
b) Who earns what share of revenues stemming from IP (patents, trademarks, design rights, etc.) created from publicly funded research in PRIs?	*Please indicate the relative percentage (confirm that the percentages add up to 100%)*	*PRIs set schemes themselves*	**Source:**
PRI			
Research unit / laboratory within PRI			
Researchers			
Other			
c) *From 2005-17, were any reforms introduced that affected the institutional autonomy of HEIs and PRIs?*	*Please indicate which reform(s)*	*No changes made*	**Source:**
Q.3.8. *Which reforms to institutional autonomy have been important to enhance the impacts of public research?*			

Comments: Indicate if any of the answers above are difficult to provide and, if so, why. Please describe how much scope HEIs/PRIs themselves have with regard to defining these conditions.

A.3. Taxonomy

Budget allocations are direct transfers to HEIs and PRIs once the public budget has been approved and transferred to the responsible ministries and/or funding agencies.

Funding instruments are specific mechanisms to allocate public research funding to performers or groups of performers. The identification of funding instruments is principally based on the following criteria: Their visibility as a unique funding instrument for research, like a clearly distinguishable programme managed by a research funding organisation; the internal consistency in terms of the allocation mechanism and criteria, as well as concerning the allocation mode (see project-based or institutional funding).

Higher education institutions (HEIs) are organisations whose main mission is to offer education at the tertiary level (ISCED levels 5 to 8), as well as to perform R&D. HEIs are generally funded through a core State attribution – institutional funding that covers generally jointly funding for education and research. Recently, higher education institutions have been adding commercialisation and support of socio-economic development (e.g. innovation) as a third mission.

Institutional autonomy refers to the capability for institutions to take independent action, in other words being self-governed.

Institutional funding means funding attributed to HEIs or PRIs for their running activities and, usually, for an unlimited period (the yearly amount might vary). Institutional funding is usually not earmarked to specific activities and to organisational subunits, but the internal allocation is left to the performing institution (e.g., block grant to public higher education institutions). These funding streams may also be performance based but not competitive.

Inter-agency joint programming refers to formal arrangements that result in joint action by implementing agencies, such as e.g. sectoral funding programmes or other joint policy instrument initiatives between funding agencies.

Performance contracts define goals agreed between ministry/agency and HEIs/PRIs and link it to the institutional funding of HEIs and PRIs.

Public research institutions (PRIs) are public-sector organisations which perform R&D activities as one of their core missions. PRIs can be broadly characterized in five types:

- *Associations of public research organizations* ("Umbrella public research organizations", UPROs, e.g., Centre National de la Recherche Scientifique CNRS in France, Fraunhofer-Gesellschaft and Max-Planck Gesellschaft in Germany, National Institutes of Health in the U.S.) are national-level organisations with the mission of organising research activities in a specific domain. They mostly host research laboratories distributed over the whole national territory and they are delegated by the State to manage a specific field of national research policy. UPROs in many cases have a dual function, i.e. to directly manage laboratories and scientists' careers on the one hand, and to provide competitive projects funds on the other hand.
- Traditional mission-oriented centres (MOCs) that are government owned and in some cases run by government departments or ministries at the national and sub-national levels. Their role is to undertake research in specific topics or sectors in order to provide knowledge and technological capabilities to support policy-making.

- *Public research centres and councils* (PRCs) perform (and in some cases fund) basic and applied research in several fields. These overarching institutions tend to be of considerable size in several countries representing a significant share of the national R&D capabilities.
- *Research technology organisations* (RTOs), also known as industrial research institutes, are mainly dedicated to the development and transfer of science and technology to the private sector and society. Although some of them are owned by government, in general, the administrative links of RTOs with governments tend to be looser than those of the other types of PRIs. RTOs often operate in the semi-public sphere and in the non-profit sector or even in the business enterprise sector (NACE 72) according to the Frascati manual (OECD, 2015).
- *Independent research institutes* (IRIs) perform basic and applied research focused on specific "issues" or "problems" (e.g. societal challenges such as mobility) rather than just fields. In many cases IRIs may be termed as "semi-public" as they are founded under different legal forms and work at the boundaries between public and private, but develop their activities with substantial public support and/or participation of public representatives in their governance.

Public institution: an institution is classified as public if it is controlled and managed directly by a public education authority or agency, or is controlled and managed either by a government agency directly or by a governing body (Council, Committee etc.), most of whose members are appointed by a public authority or elected by public franchise.

Private institution: an institution is classified as private if it is controlled and managed by a non-governmental organisation (e.g. a Church, Trade Union or business enterprise), or with a Governing Board consisting mostly of members not selected by a public agency.

Project-based funding are funds that are attributed to a group or an individual to perform a research activity limited in scope, budget and time. Project-based funding can be distinguished from institutional funding based on three main characteristics: a) funds are attributed directly to research groups and not to a whole organisation; b) funds are limited in the scope of the research supported and its duration; and c) funds are attributed by a research funding organisation outside the performing organisation to which the beneficiary belongs. Funding for research supports research personnel, basic and applied research activities, research infrastructure and/or equipment. Funding for innovation supports technology development, technology transfer, research commercialisation, and/or joint R&D between PRIs and industry.

Research and innovation council/committee is any non-temporary public body that takes decisions concerning HEI and PRI policy, and that has explicit mandates by law or in its statutes to either provide policy advice (i.e. produce reports); oversee policy evaluation; coordinate policy areas relevant to public research (e.g. across ministries and agencies); set policy priorities (i.e. strategy development, policy guidelines); and/or joint policy planning (e.g. joint cross-ministry preparation of budgetary allocations).